

There Is No Planet B: Aligning Stakeholder Interests to Preserve the Amazon Rainforest

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Received: November 28, 2020

Revised: September 12, 2021; February 4, 2023

Accepted: March 29, 2023

Published Online in *Articles in Advance*: August 23, 2023

<https://doi.org/10.1287/mnsc.2023.4884>

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Abstract. How do firms address complex collective action problems effectively? Institutional and stakeholder research suggests that firms may avoid the tragedy of the commons by aligning the interests of critical proximate stakeholders in ways that governments cannot accomplish. This phenomenological paper investigates this possibility by analyzing Amazon rainforest preservation by Natura, a Brazilian cosmetics company. The results indicate that Natura internalized environmental externalities by linking ecologically conscious consumers with rural Amazonian communities. A differences-in-differences analysis compares forest preservation and fire activity in the municipalities that Natura entered with those in which it did not enter. Natura's impact is identified through an instrumental variable analysis using missing satellite images, which Natura relied upon to decide which municipalities to enter. Quantitative results tie Natura's entry into municipalities with forest preservation. Analysis of three mechanisms associates Natura's involvement with stakeholder decisions to cultivate diverse forest-generated crops instead of clearing the land for conventional agriculture. This study contributes to the management literature by suggesting how firms can address important global challenges, such as rainforest preservation, by investing in stakeholder capability development and by creating institutional arrangements in line with those envisioned elsewhere.

History: Accepted by George Serafeim, special section of management science: business and climate change.

Funding: This work was supported by the Clarkson Centre for Business Ethics [CAD 7,500.00] and Canada's Social Sciences and Humanities Research Council.

Supplemental Material: The data files and online appendices are available at <https://doi.org/10.1287/mnsc.2023.4884>.

Keywords: Coasean arrangements • new stakeholder management • satellite imagery • Amazon rainforest • sustainable development goals (SDGs)

Introduction

Climate change is a grand challenge of the highest order (George et al. 2016, Henderson and Serafeim 2020, Serafeim 2022). As a common good, the natural environment has been excessively exploited and depleted in a "tragedy of the commons" (Hardin 1968; Ostrom 1990, 2005). This challenge and tragedy have led management scholars to call for research on causes and remedies (Bansal and Roth 2000, Dowell et al. 2000, Delmas and Toffel 2008, Lyon and Maxwell 2011, Bansal and DesJardine 2014, Henderson et al. 2015, Luo and Kaul 2019, Nardi 2021). The new stakeholder management literature responds to this call by suggesting that the alignment of stakeholder interests by a firm may address collective-action problems such as climate change more effectively than can occur through governments in some situations (Porter and Kramer 2006; Barney 2018; Bridoux and

Stoelhorst 2020; McGahan 2021, 2023; Bacq and Aguilera 2022; Gatignon and Capron 2023). However, despite the potential for corporate action in the public interest, relatively little is known about how firms manage their stakeholders to address complex collective action problems such as environmental preservation (Odziemkowska and Dorobantu 2021).

The contribution of this phenomenological paper is the elucidation of the impact on the Amazon rainforest of an innovative Coasean arrangement implemented by Natura, a Brazilian cosmetics company. Founded in 1969, Natura in 2021 was the world's largest B-Corporation with net revenue of R\$40.2 billion (equivalent to US\$7.2 billion).¹ One of its distinctive characteristics is a commitment to positive environmental impact (Boehe et al. 2014, Nar-salay et al. 2015, Lazzarini et al. 2020a, Marquis 2020, Gatignon and Capron 2023). A primary practice that Natura

implemented to advance its environmental mission is procurement of ingredients from the Amazon rainforest to produce a diverse range of cosmetics, shampoos, and soaps within the *Natura Ekos* product line.² Recognized as a purpose-driven corporation (Gartenberg et al. 2019, Gartenberg and Serafeim 2022, Serafeim 2022), Natura received the United Nations' Champion of the Earth Award in 2015.³

The analyses reported in this paper assess the efficacy of Natura in aligning stakeholder interests to resolve common-goods problems by constructing arrangements through which environmentally conscious consumers pay premium prices for branded products that are manufactured with ingredients that originate with Indigenous suppliers in the Amazon region. By aligning the interests of proximate stakeholders with its own interests, the firm simultaneously achieves both financial and environmental aims (Delmas and Toffel 2008, Devinney et al. 2013, Henderson et al. 2015, Bowen et al. 2018, Lumpkin and Bacq 2019). This process involves more than reducing transaction costs in a narrow sense. As discussed below, Natura and community leaders work together to establish enduring relationships and capabilities that support sustained exchange. We refer to these as "Coasean arrangements" rather than as "Coasean bargaining" because of their durability and complexity. A primary contribution of this research is to raise questions for theory regarding the commitments required of firms that seek to implement Coasean arrangements to resolve common-goods problems such as exploitation of the environment.

A second contribution is the empirical testing of Coasean arrangements to address climate change. The quantitative analyses reported in this paper focus specifically on how Natura's entry into Amazonian communities internalized the externalities of forest preservation. Natura's impact is assessed through a differences-in-differences estimation model with two-way fixed effects (Chakraborty and Jayaraman 2019, de Chaisemartin and D'Haultfoeuille 2020, Callaway and Sant'Anna 2021, Goodman-Bacon 2021). Rainforest preservation in the Brazilian Amazon between 2000 and 2018 is measured using publicly available georeferenced satellite data as advocated by Nagaraj (2022) and Ostrom and Nagendra (2006) on the number of forested hectares that survive over time and on the number of fire incidents per municipality. The data analysis includes an instrumented differences-in-differences estimation model (Mariano 1977, Hudson et al. 2017, Nagaraj 2022), which identifies Natura's impact through missing images from the satellites that Natura's managers use in entry decisions. Through the analysis of mechanisms, we establish that Natura's entry is associated with greater levels of Amazonian rather than conventional agriculture; that Natura's involvement is tied to greater Amazonian agricultural heterogeneity; and that Natura tends to enter old-growth areas with high

carbon mass, where it appears to mitigate fire vulnerability. The findings are integrated with qualitative assessments developed from field visits and interviews (Boehe et al. 2014, Lazzarini et al. 2020b).

Third, the paper also integrates into the field of management considerations of performance that reflect the United Nations' sustainable development goals (SDGs).⁴ Overall, Natura and its stakeholders are identified with the preservation of 730,000 hectares of the Amazon during the two decades under study. This area is equivalent to 676,000 football (a.k.a. soccer) fields.⁵ The mechanisms giving rise to this result involve collective decisions executed by Natura, environmentally conscious customers, community leaders, and local Indigenous farmers. Preservation occurred as farmers in municipalities vulnerable to rainforest clearing actively chose to harvest fruits, nuts, seeds, and berries. By aligning the economic interests of these proximate stakeholders, Natura compelled Indigenous farmers not to choose conventional agriculture, such as soybean and cattle farming, that requires clearing land through deforestation. Instead, stakeholders jointly elected to pursue the cultivation of forest-compatible products such as açai and cocoa, among other Amazonian products. The heterogeneity of the Amazonian products sourced by Natura from local farmers drove both forest conservation and forest regeneration. The environmental impact of Natura's entry into Amazonian municipalities was scaled, enduring, and significant, and thus serves as an example of how a private-sector organization can advance public aims such as the reduction of climate change.

Stakeholder Management Promoting Coasean Solutions to Common-Goods Problems

The present study draws on two main streams of literature: *institutional economics* (Coase 1960; Demsetz 1967; Hardin 1968; Libecap 1989; North 1990; Ostrom 1990, 2000; Williamson 1996; Stavins 2011) and *new stakeholder management* (Donaldson and Preston 1995; Delmas and Toffel 2008; Devinney et al. 2013; Henisz et al. 2014; Dorobantu and Odziemkowska 2016; Flammer and Kacperczyk 2016; Barney 2018; Jones et al. 2018; Klein et al. 2019; Lumpkin and Bacq 2019; Bridoux and Stoelhorst 2020; McGahan 2021, 2023; Stoelhorst 2021; Bacq and Aguilera 2022). Both suggest that private-sector action may address common-pool problems such as the exploitation of natural resources. This section briefly describes these two streams to highlight the puzzle that this study addresses.

Tragedy of the Commons and Potential for Coasean Solutions

In general, joint value-creation problems among transacting actors can take on the character of common-goods

problems. This occurs particularly when the parties to a transaction face intertemporal inconsistencies in their incentives to invest (Mahoney et al. 2009, Bansal and DesJardine 2014, Klein et al. 2019). Ostrom's seminal work describes the challenges of establishing rules and norms to govern common-pool resources (Ostrom 1990). Governing the commons is complicated because, by definition, relevant resources are rivalrous in consumption and are nonexcludable in access or use. These features lead to ill-defined property rights, weak incentives for preservation, and consequently to the "tragedy of the commons" (Demsetz 1967, Hardin 1968, Libecap 1989, Stavins 2011). The tragedy is that stakeholders each have private incentives to exploit the resource to the detriment of the common interest. This malalignment can lead to the overexploitation of common-pool resources to their exhaustive depletion (Demsetz 1967, Hardin 1968, Libecap 1989, Ostrom 1990, Stavins 2011, Marciano et al. 2019).

The prevalence of the tragedy of the commons in the environment is extensive and includes detrimental exploitation of fishery waters, river basins, air pollution, communal grazing lands, and natural forests. In each of these environments, access to resources is nonexcludable among a proximate group of stakeholders. Open access among stakeholders with an incentive for private usage leads to a negative externality for the group. In the case of natural forests, where property rights are held jointly, individual stakeholders have logged, mined, trafficked wild animals, and deforested them for private agricultural use. In doing so, these stakeholders have generated private gains and public losses (Demsetz 1967, Hardin 1968, Libecap 1989, Ostrom 1990, Ostrom and Nagendra 2006, Stavins 2011).

Some of the mechanisms used by governments to remediate the private appropriation of common-pool resources are laws and regulations about the use of common-pool resources (Stigler 1971, Crocker and Masten 1996), taxations and penalties for deviant behavior among agents (Pigou 1929), and the use of coercive power through command and control operations (Weber 2004, Souza-Rodrigues 2019). However, these responses can be ineffective, particularly when governments do not have adequate resources and capabilities (Cabral et al. 2019, Lazzarini et al. 2020b), must attend to competing aims that may dilute attention to environmental problems (Alston et al. 1999, 2000; Symonds 2019), and/or are instrumentally led by actors who see no direct benefit from the engagement of stakeholders in joint value creation (Williamson 1999, Bridoux and Stoelhorst 2020, Bacq and Aguilera 2022).

Coase (1960) argued that an alternative institutional approach using the market mechanism might be feasible under certain conditions. The argument is that, if the transaction costs of using the market mechanism are low enough, then production value will be higher after

agents freely negotiate and rearrange their legal rights over resources and the production externalities. In other words, in a situation of low transaction costs, the market mechanism can be developed to internalize externalities and increase production value through Coasean bargaining among the participants (Coase 1960). The bargaining itself is referred to as "private ordering" in this literature (North 1990). Coase (1960) describes how, through private ordering, the private parties transfer value between them such that positive externalities are internalized. The result, Coase (1960) argues, can be a more efficient and effective arrangement than what could be accomplished by the government. Coase (1960) assumes that it is straightforward to identify stakeholders that are directly affected by the externality and assumes that negotiations among stakeholders are feasible because all understand the nature of the common problem and have the sufficient capacity to negotiate.

In the case of the Amazon rainforest, neither of the Coasean assumptions is robust as Indigenous farmers often have had no contact with corporations or experience in market exchange prior to interacting with Natura. In such a situation, some policymakers and scholars might suggest that the government can and should intervene to facilitate bargaining between agents as an alternative to enacting laws and regulations. However, in this instance, the government did not intervene, and the private institution acted instead to implement a Coasean arrangement. Transaction costs in the sense identified by Coase (1960) are one form of impediment to the achievement of this alignment. However, other impediments also arise, such as a lack of contact among relevant stakeholders and the absence of market-engagement capacity by some stakeholders. The essence of the Coasean insight on which we draw is that a private actor such as a corporation may achieve an alignment of interests through the private ordering of stakeholder interests. The critical idea from Coase (1960) is that stakeholder decisions not to exploit or to restore common-pool resources, such as natural forests, may be supported by a firm that overcomes these impediments to align stakeholder interests in ways that lead to the internalization of environmental externalities.

Coase (1960) provides several examples of how Coasean bargaining might be Pareto efficient to internalize environmental externalities when transaction costs are negligible. For instance, in the case of a polluting factory, Coase demonstrates that if the factory owner and the neighboring property owners can freely bargain, they may achieve a more efficient allocation of rights and constraints than a government seeking social welfare. In this example, the private stakeholders internalize pollution's negative externalities and deploy the market to allocate resources more efficiently than can a regulator. In the case of the Amazon rainforest, the remediation of negative externalities occurs with the implementation of

arrangements that prevent clear-cutting and burning. As shown later, this is the primary source of Natura's impact.

A parallel logic applies to the internalization of positive externalities. For instance, suppose an agent cultivates a rose garden in a public space within a neighborhood. If transaction costs are minimal, the agent may bargain with other residents living in neighboring properties aiming to obtain rights and resources for the garden in exchange for pleasing scents and gracious visual experiences. As also shown later, Natura implements arrangements that constitute the achievement of positive externalities through the reforestation of previously clear-cut areas analogously to the rose garden.

In sum, firms and their proximate stakeholders may implement Coasean arrangements to internalize environmental externalities by achieving an alignment of mutual interests. Although this may be accomplished, as Coase (1960) emphasized through the reduction of transaction costs through bargaining in an established market, the broader insight is that, even where markets do not yet exist and market-making capacities of some stakeholders have not developed, private actors can work together to align interests, overcome impediments, and internalize externalities. We turn now to the *new stakeholder management* literature to understand how a firm might establish these arrangements.

Stakeholder-Oriented Firms as Drivers of Coasean Arrangements

Recent advances in the *new stakeholder management* literature indicate that firms with a purposeful stakeholder orientation have several advantages over firms with a shareholder orientation (Barney 2018, Gartenberg et al. 2019, Kaplan 2019, McGahan 2021, Gartenberg and Serafeim 2022, Serafeim 2022). In some situations, an orientation toward stakeholders may immediately and directly improve a firm's economic performance (Jones 1995; Flammer 2013, 2015; Eccles et al. 2014; Henisz et al. 2014; Barney 2019; Gartenberg et al. 2019; Nardi et al. 2022), as well as its environmental performance (Bansal and Roth 2000; Delmas and Toffel 2004, 2008; Devinney et al. 2013; Henderson et al. 2015; Bowen et al. 2018; Lumpkin and Bacq 2019). In a different class of situations, the achievement of a stakeholder orientation requires innovation in the enfranchisement and alignment of stakeholders with the firm through a reconstruction of the collective's approach to value creation, appropriation, and distribution (Donaldson and Preston 1995; Donaldson 2002; Freeman et al. 2004; Porter and Kramer 2006, 2011; Garcia-Castro and Aguilera 2015; Lieberman et al. 2017; Horwitz and McGahan 2019; Kaplan 2019; Klein et al. 2019; Amis et al. 2020; Bridoux and Stoelhorst 2020; McGahan 2020; Stoelhorst 2021; Bacq and Aguilera 2022).

In this second situation, stakeholder-oriented firms can reduce transaction costs, align interests among diverse agents, resolve intertemporal inconsistencies, equalize negotiation capabilities, and provide innovative solutions to collective problems (Hart and Dowell 2011, Bansal and DesJardine 2014, Eccles et al. 2014, Luo and Kaul 2019). This literature suggests that a purpose-driven private corporation (Serafeim 2022) may be able to resolve collective action problems more effectively than the government by extending Coasean arrangements in specific ways that are inaccessible to the government. In particular, a private corporation may have the capacity to cultivate negotiation capabilities in local stakeholders with control over intertemporally valuable resources. Through training programs, long-term contracting, and relationship commitments, a company may cultivate institutional capabilities in previously under-supported stakeholders that ultimately lead to the development of sustainable markets (Bansal and DesJardine 2014; Eccles et al. 2014; Luo and Kaul 2019; McGahan 2021, 2023; Gatignon and Capron 2023). Such investments are justified because they yield private benefits to the corporation over the long term. The logic driving this arrangement extends the conceptualization of Coasean solutions beyond market-creation and institutional reform into the targeted development of trading partners in market-based skills and the mutual development of an institutional environment that enables the sustainable operation of markets (North 1990).

Gatignon and Capron (2023) showed that Natura enacted these and other types of solutions in its interactions with diverse actors such as rural suppliers, civil society organizations, direct sellers, shareholders, and final consumers. The firm addressed failures in local market institutions by developing an "open institutional infrastructure," which reduced local transaction costs and empowered a wide range of actors, including both participants and nonparticipants in direct exchange with Natura (Gatignon and Capron 2023). Natura's enablement of an efficient system for sourcing Amazonian agricultural products increased its own profitability by shifting volume toward products that commanded premium prices because of their Amazonian content, which led, in turn, to the allocation of financial value to Indigenous suppliers of ingredients (Boehe et al. 2014, Narsalay et al. 2015, Lazzarini et al. 2020a, Marquis 2020, Gatignon and Capron 2023).

The following analysis shows the effects of the Coasean arrangement developed by Natura to overcome impediments that had prevented links between environmentally conscientious consumers and rural Amazonian communities. The distribution of value to local Amazonian communities aligned the interests of local community members with the preservation of the forest for the cultivation of fruits, nuts, seeds, and berries that were subsequently sold to Natura and incorporated into its

premium-priced, branded products. Through a value-sharing system, Natura and its proximate stakeholders internalized the negative environmental externalities that would otherwise occur through clear-cutting and/or burning for conventional agriculture. As noted above, Natura also sought to reforest in previously cleared areas, which constitutes internationalization of the positive environmental externality of recreation of the natural environment, although this step is less prevalent than the deterrence of deforestation.⁶

Natura and the Amazon Rainforest Communities

The development of an ecosystem of suppliers in Amazon communities arose gradually as Natura sought to develop flexible contracts and trusting relationships with local Indigenous communities in municipalities rather than procuring conventional products through arms-length transactions. Over time, these relationships deepened as Natura collaborated with local leaders to engage in farmer training and other supporting activities. Importantly, Natura's contracts did not bind communities as exclusive suppliers (Lazzarini et al. 2020a) and could be severed by a community at any time, although there is no instance in the data set of a municipality ceasing to source ingredients to Natura after commencing arrangements.

Historically, the Amazon region has been neglected by both the private and the public sectors. For decades, the region was considered the archetype of wilderness and underdevelopment (Fernández-Llamazares et al. 2020). Some political figures in Brazil went as far as to claim that the natural environment in the Amazonian region impeded the country's economic progress and that the rainforest should be destroyed (Pokorny et al. 2012, 2013; Nazareno and Laurance 2020). Under this framework, deforestation was encouraged to yield economically valuable timber and animals (Pokorny et al. 2013, Pokorny and Pacheco 2014, Nunes et al. 2020, Rajão et al. 2020, Silva et al. 2020). The voracious depletion of these natural resources in the region was accompanied by human rights violations, including underpayments, enslavement, and violence against native persons (Mendes et al. 1989, Alston et al. 2000).⁷

To establish a network of Amazonian suppliers, Natura invested during the 2000s in its "Amazon Plan," which increased its presence from 4 municipalities in 2000 to 56 municipalities in 2018 (please see Online Appendix 2 for a visual representation of this expansion).⁸ To implement the "Amazon Plan," Natura performed several activities. Aiming to be energy efficient and pool production at a single location (Plambeck 2012), the firm built an industrial plant in the Amazon (city of Benevides, Pará State), which subsequently became an "Eco-Park" for multiple firms working in the Amazon region. Natura established a dedicated team to develop relationships with rural Indigenous

communities called "Ecorelations," which was crucial to establishing deep relationships with Indigenous Amazonian community leaders through extensive visits during which Ecorelations staff and local leaders built mutual trust and personal connections prior to discussing the details of contractual arrangements with implications for community farmers. Many Amazonian stakeholders had no familiarity with the legal system in which contracts are written and enforced prior to interaction with Natura. Several had never engaged in commercial relations with a firm at all. For most, the ability to bear intertemporal financial risk is minimal as many had not held currency or a bank account. An essential feature of the arrangements was the ability of Natura's Ecorelations staff, many of whom were members of Indigenous communities themselves, to understand that, in Indigenous Amazonian cultures, value is conceptualized in ways that are fundamentally different than implied by utilitarian theories of economics.

The mandate of the Ecorelations team was to explore mutual interests with Indigenous community leaders to source Amazonian ingredients for new cosmetics, shampoos, and soaps. After identifying a potential ingredient with technical and marketing appeal, the team began discussions with community leaders in municipalities regarding potential arrangements. An important step toward writing a contract was often the establishment of an association or cooperative in the locality, and for collectively setting the terms of sale on price and quantity. Members of Natura's Ecorelations team worked with communities to set up bank accounts and to implement arrangements for transporting products to the Eco-Park. This process was followed, legitimized, and supported by local nongovernmental organizations (NGOs), grass-roots organizations representing Indigenous groups, and civil society associations that together constituted Natura's stakeholder-oriented governance system (Boehe et al. 2014, Narsalay et al. 2015, Lazzarini et al. 2020a, Marquis 2020, Gatignon and Capron 2023).

After arrangements were established with Indigenous community leaders on behalf of local municipalities, Natura's team worked with the community to identify processes through which ingredients would be extracted without damaging the forest to ensure the forest's sustainability over time (Saes et al. 2014). A critical feature of this approach was the sustainable harvesting of Amazonian ingredients from the forest into perpetuity. Rural suppliers then became responsible for collecting, processing, and shipping the Amazonian products to Natura's Eco-Park, where they received payment for the ingredients they delivered (please see Online Appendix 3 for photos from the field). At the plant, the ingredients were transformed into natural oils, butters, and pastes that became the essence of the cosmetics, shampoos, and soaps in the *Natura Ekos* product line. Natura advertised the products' social, environmental, and technical benefits to final consumers (please see Online Appendix 4).

Natura's payments to Amazonian municipalities for fruits, nuts, seeds, and berries were set to convey sufficient value as to preserve the alignment of supplier interests with rainforest preservation. The absence of a market for these fruits, nuts, seeds, and berries prior to Natura's arrival and the absence of market-making experiences within the Indigenous communities created challenges that Natura sought to address. Figure 1 shows that, among the total payments that Natura made to the communities, on average, 39% were direct payments for Amazonian products supply. The remaining 61% were for additional benefits to the community.⁹ These benefits were distributed in several different ways. For example, Natura made payments for access and use of traditional knowledge (71.5% of the total), local research on Amazonian products (16.5% of the total), investments in infrastructure and institutional support (5% of the total), provision of training programs and technical assistance (4% of the total), and compensation for commercial images and carbon credits (3% of the total). In 2019, Natura paid Amazonian suppliers a total of R\$33.5 million (US\$8.33 million as of December 31, 2019), of which R\$20 million (US\$5.1 million as of December 31, 2019) were additional benefits over payments for the products themselves.¹⁰

Natura employees, executives, and shareholders emphasized the importance of sharing value with local communities to align interests and maintain Amazonian supplies. The distribution of value enabled the alignment between environmental and economic interests within the Amazonian rural communities. According to Natura's chief executive officer (CEO):

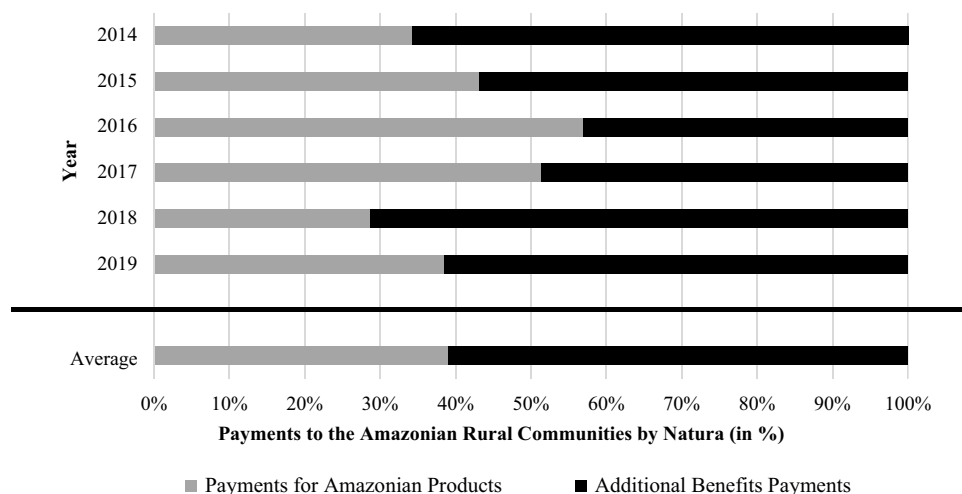
"To prevent ucuúba trees from being cut down for the production of brooms, sold in local shops, the group [Natura] decided to pay twice the amount that the Amazon community received for their work.

Instead of cutting, they started to earn more to keep the trees upright and extract only the seeds, which have moisturizing properties for the production of creams." Roberto Marques, Natura Co. CEO. Available at UOL ECOA.¹¹

Natura benefitted from these Coasean arrangements directly in several ways, including through higher profitability than the estimated average for global cosmetics companies (Boehe et al. 2014).¹²

Natura's value-sharing approach has some features that are similar to the Fairtrade model. Both models focus on paying premium prices to the suppliers of differentiated inputs. These premium prices support the economic development of supplying communities while aligning interests to preserve the local environment, engage in training, advance local leadership development programs, and strengthen democratic institutions (Saes 2009, Nelson and Pound 2009, Haggard 2013, Darko et al. 2017, Fenger et al. 2017, Silva et al. 2020, Loconto et al. 2021). However, some differences also arise between Natura's approach and the Fairtrade approach. In particular, Natura enters into Amazonian communities that may have no (or minimal) prior contact of any kind with currency, bank accounts, commercial production, contracts, or large-scale market exchange. Natura's model of engagement, which is akin to polycentric governance (Ostrom 1990, Gatignon and Capron 2023), involves mutual decision making between the firm and Indigenous communities to discuss and decide production quantities, prices, and additional investments required to sustain the arrangements (Boehe et al. 2014). The emphasis is on collective deliberation between Natura and Indigenous community leaders, who in turn worked closely with rural community members according to traditional norms (Boehe et al. 2014). These

Figure 1. Payments for Amazonian Products and Additional Benefits Payment by Natura to the Amazonian Rural Communities



Notes. Data from Natura's Annual Report 2019 and Annual Report 2016. GRI 203-1 indicator. Both were accessed on May 28, 2020. More details are presented in Online Appendices 11 and 12.

features of Natura’s approach suggest that it internalizes externalities through long-term arrangements that involve capability-building among future trading partners that ultimately enable stakeholder interest alignment (Luo and Kaul 2019, Dowell et al. 2000, Teodorovicz et al. 2023).

Data and Method

The purpose of the empirical analysis is to seek evidence regarding whether and how Natura operated in the Amazon to remediate negative externalities through the deterrence of deforestation and to support positive externalities through reforestation.

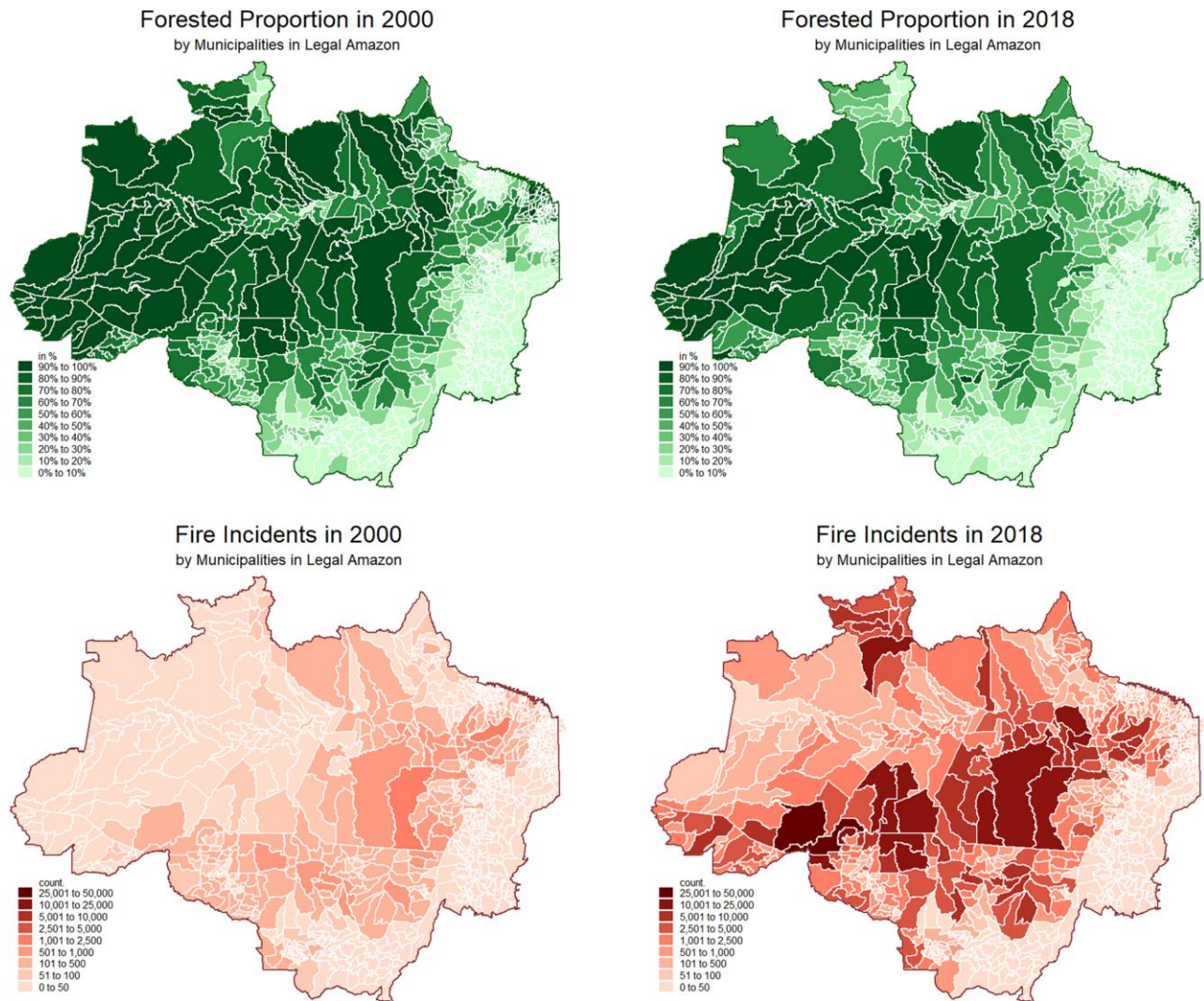
Data and Variable Definitions

The main database on which the analysis depends is drawn from the Brazilian National Institute for Spatial

Research (INPE), which reports georeferenced satellite information on environmental protection for each of the 760 municipalities in the Brazilian Amazon region from 2000 to 2018. The INPE system uses images from Landsat (NASA) satellites, which have a spatial resolution of 30 m. This system has been internationally recognized as a reliable monitoring system for forest preservation (Kintisch 2007). Figure 2 presents geographic information system (GIS) mapping with data from the first and last year in our panel.

INPE’s database was merged with a data set made available from Natura, which specifies the geographical locations of each Amazon supplier, the year in which Natura entered those locations, and the Amazon products that these suppliers provide. Finally, we use data from the Brazilian Geography and Statistics Institute (IBGE) on the yearly Municipal Agricultural Production census.

Figure 2. (Color online) Forested Proportion and Fire Incidents in Brazilian Amazon from 2000 to 2018



Notes. Data from INPE PRODES, Terra Brasilis, and IBGE. Considering an open lower bound in the legend categories of forest proportion maps.

Table 1. Descriptive Statistics

No.	Variable	Observations	Mean	Standard deviation	Minimum	Maximum	Correlation matrix		
							1	2	3
1	1(Natura)	14,440	0.04	0.19	0.00	1.00	1.000		
2	Log (Forested Area + 1)	14,440	5.12	3.25	0.00	11.93	0.121**	1.000	
3	Log (Fire Incidents + 1)	14,440	3.95	2.95	0.00	11.43	0.149**	0.644**	1.000

** $p < 0.01$; * $p < 0.05$; + $p < 0.1$.

Dependent Variables. To measure the degree of environmental protection and depletion, we use two primary outcomes. The first, *Forested Areas* (in km²), describes the total preserved area of the Amazon rainforest in each municipality. This value is updated every year by INPE's satellite images. The second, *Fire Incidents* (in number of occurrences), shows the number of fire incidents in each year by municipality. Unfortunately, INPE does not differentiate wildfire occurrences from anthropic fire occurrences. Nevertheless, given the logic of the Amazonian deforestation process, these fire incidents are usually caused by farmers and loggers aiming to “clear” forested areas for future cultivation and cattle farming (Alston et al. 2000, Pokorny et al. 2013, Costa 2020, Rajão et al. 2020, Silva et al. 2020). We proceed with caution in this instance and interpret the results as only suggestive as a result of these features of the data and of the inability to observe directly why and how Natura acts in these areas. We linearize both variables given the skew in their distribution (Galasso and Schankerman 2015).¹³

Independent Variable (Natura). The primary independent variable is a binary measure that takes the value of one if Natura entered the municipality, that is, the firm established supply relations with a rural community in the municipality in that particular year, and zero otherwise. This variable is not colinear with yearly effects because Natura entered municipalities successively over the 19 years under analysis. This variable is characterized as

$$Natura_{m,t} = \begin{cases} 1 & \text{if } m = \text{Natura Municipality and } t = \text{Post Entrance} \\ 0 & \text{if } m = \text{Not a Natura Municipality or } t = \text{Pre Entrance.} \end{cases}$$

Controls. As both dependent variables rely on satellite images, we include yearly satellite controls, including information on the amount of “not forested areas” (in km²), which are natural savanna areas or urban areas; “hydrographic areas” (in km²), such as lakes, rivers, and sea; “cloud areas” (in km²), that is, areas in which the satellite system does not discern topographic characteristics because of cloud coverage; and “not observed areas” (in km²), that is, areas that the satellite could not record the

topographic coverage. Control variables may vary by municipality and year.

Satellite images are considered an ideal measure for tracking forest evolution among researchers studying the tragedy of the commons. According to Ostrom and Nagendra (2006, p. 19225), “Satellite remote sensing is the most frequently used technique for mapping changes in forest cover. When combined with on-the-ground observations, studies of land-cover change enable us to analyze social incentives and actions and explore environmental and social change.” Table 1 presents basic descriptive statistics.

Estimation Models and Results

We expect Natura's entry into a municipality to impact forest preservation positively compared with what would have occurred had Natura not entered.¹⁴ To test this proposition, the paper reports on basic descriptive analysis applying a two-way fixed effect differences-in-differences analysis (TWFE DiD) (Chakraborty and Jayaraman 2019, de Chaisemartin and D'Haultfoeuille 2020, Callaway and Sant'Anna 2021, Goodman-Bacon 2021). However, because Natura chose municipalities for entry, there is fundamental endogeneity in selection. To address this, we perform an instrumental variable differences-in-differences analysis (DiD-IV) (Mariano 1977, Angrist and Pischke 2008, Hudson et al. 2017, Nagaraj 2022). The crucial difference between the unadjusted (TWFE DiD) and the adjusted (DiD-IV) models is the correction in the DiD-IV model for Natura's selection. The benefit of performing both estimations is assessing whether this selection is associated with any specific heterogeneous trends between treated and control municipalities prior to the firm entry.

TWFE DID

The first estimation model (TWFE DiD) enables verification of the effect of Natura's entrance β on forestation and the number of fire incidents in a municipality m subsequently from the year of entry $t = 0$. Municipality fixed effects are represented by γ_m , and year fixed effects are represented by λ_t . The model

$$Y_{m,t} = \alpha + \gamma_m + \lambda_t + \beta \text{Natura}_{m,t} + X'_{m,t} \rho + \epsilon_{m,t}$$

assumption behind the TWFE DiD model is that treated and untreated units would have parallel trends in the

absence of the treatment. In other words, if Natura had not entered, municipalities would similarly tend toward deforestation and fire incidents, that is, $E[Y_{m,t} | m, t] = \gamma_m + \lambda_t$.

Results

The first set of results provides evidence that Natura’s entry into a municipality is positively tied to forest preservation in that municipality. As indicated in the TWFE DiD model presented in Table 2, column (2), forest preservation is 47.9% ($p < 0.05$) higher in municipalities where Natura entered. According to the TWFE-DiD estimation, Natura’s entry in the municipalities contributed to the preservation of about 1.8 million hectares, which is equivalent to 1.7 million football (a.k.a. soccer) fields, relative to what the model estimates would have occurred had Natura not entered.¹⁵ Figure 3 is an event study-style graph (Clarke and Schythe 2021) showing that, from the year that Natura enters onward ($t \geq 0$ year indexes), there is a positive trend toward forest preservation. The trend is significant both in cross-section and intertemporally, as the trend lines after Natura’s entry are statistically different from those prior to Natura’s entry.

The results on fire incidents are complex. Although Table 2, column (4), shows that the number of fire incidents in the municipalities that Natura entered increased by 49.4% after Natura’s entry ($p < 0.01$; equivalent to 1,100 total fires), Figure 3 suggests different pretreatment trends ($t < 0$ year indexes) between treated and control groups. Figure 3 is an event study-style graph showing the estimated effect of Natura ($t \geq 0$ year indexes). This leads to a conjecture that Natura may have elected to enter municipalities in which fire incidents were extensive and imminent, which is consistent with the finding that, even after the firm’s entry, the fire incidents increased.¹⁶ In the reported analysis, we find evidence that the increase was less than projected by the pretrend, however (please see Online Appendix 7). This suggests that Natura may have chosen to enter imminently

vulnerable fire areas deliberately. Although, without direct evidence of that choice, we leave definitive conclusions about this possibility for subsequent researchers.

DID-IV

As indicated, one of the main limitations of the TWFE DiD model is the treatment endogeneity. This endogeneity arises from the lack of randomness in Natura’s entry decision. In other words, Natura selects the municipalities for entry. This means that we cannot assess through the TWFE DiD whether the observed effects are due to Natura’s actions or if they arose independently of Natura.¹⁷ To address the endogeneity, we sought an instrument that randomly impacts Natura’s choice of municipalities but does not directly affect the topographic coverage of these locations. Such an instrument supports additional analyses using a DiD-IV (Mariano 1977, Angrist and Pischke 2008, Hudson et al. 2017, Nagaraj 2022).

Instrumental Variable (“Missing Satellite Images”)

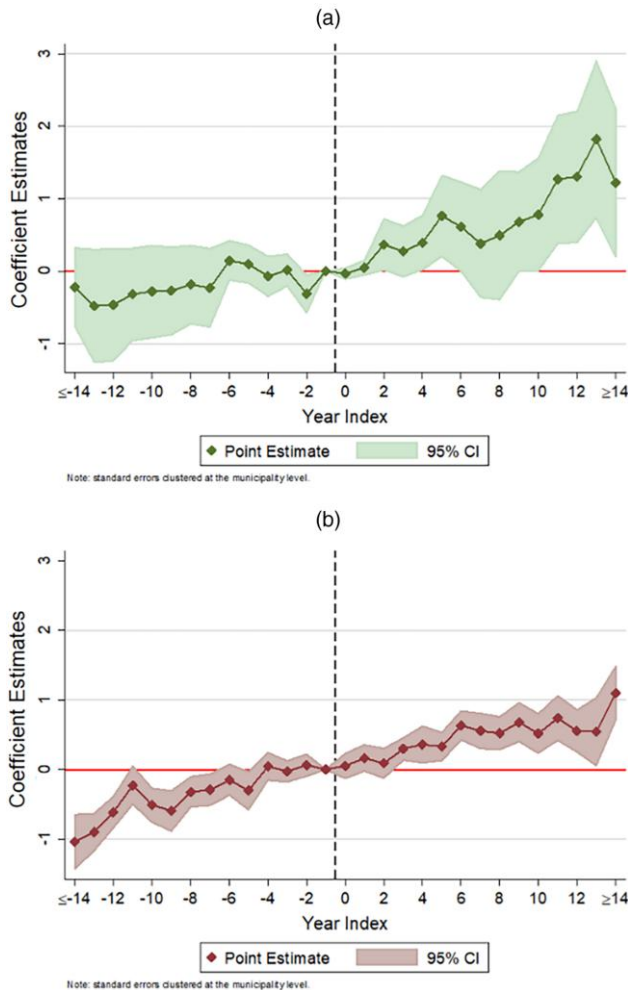
An effective instrument $Z_{m,t}$ must satisfy three assumptions. First, the instrument must be relevant, that is, $\text{corr}[Z_{m,t}, \text{Natura}_{m,t}] \neq 0$ for all municipalities m and years t . Second, the instrument must be independent, that is, unrelated to omitted variables present in the error term $Z_{m,t} \perp \epsilon_{m,t}$ or $E[Z'_{m,t}, \epsilon_{m,t}] = 0$. Finally, the instrument must satisfy the exclusion restriction, according to which the instrument $Z_{m,t}$ only affects the outcome $Y_{m,t}$ through a single channel, the treatment variable, that is, the $\text{Natura}_{m,t}$ dummy.

The use of satellite data for instrumentation has been explored in the literature, particularly in Nagaraj (2022), which uses cloud cover as an instrument for firms’ investment decisions.¹⁸ This study applies an instrument based on satellite data as well, but in this case, we consider missing satellite images. Missing images are technical omissions that arise from random occurrences with

Table 2. TWFE Differences-in-Differences Analysis of Natura Effect on Forested Areas Preservation and Fire Incidents

	Dependent variable: $\text{Log}(\text{Forested Area} + 1)$		Dependent variable: $\text{Log}(\text{Fire Incidents} + 1)$	
	(1)	(2)	(3)	(4)
Natura	0.468* (0.231)	0.479* (0.224)	0.495** (0.0850)	0.494** (0.0844)
Year fixed effects	Yes	Yes	Yes	Yes
Municipality fixed effects	Yes	Yes	Yes	Yes
Satellite controls	No	Yes	No	Yes
Constant	5.232** (0.0371)	5.303** (0.0368)	2.565** (0.0446)	2.541** (0.0449)
Observations	14,440	14,440	14,440	14,440
R ²	0.949	0.951	0.955	0.956
Number of municipalities	760	760	760	760

Note. Robust standard errors clustered at the municipality level in parentheses.
** $p < 0.01$; * $p < 0.05$; + $p < 0.1$.

Figure 3. (Color online) Evolution of Forest Preservation Indicators

Notes. (a) Forested area. (b) Fire incidents. Baseline set at year immediately before Natura's entrance. Graphs accumulate extreme periods at $t \leq -14$ and $t \geq 14$ year indexes.

satellites rather than features of the natural environment (Figure 4).

The INPE georeferencing system relies on Landsat satellites (Kintisch 2007). These satellites register images on 220 different scenes covering the entirety of the Brazilian Amazon region every year (Camara et al. 2006, p. 11). Sometimes the satellites failed to take a picture of particular scenes due to technical problems at the moment that a satellite is activated to take a picture. The failure leads to the absence of an image for a particular scene in a given year. The absences are reported as "not observed areas" within municipalities and converted to squared kilometers. In the final database, INPE registers the amount of areas for which images were missing by municipality m at year t . As described by one of INPE's coordinators:

"(...) in the past, you had fewer data available; 10 to 15 years ago, you basically had Landsat [images] only once a year and would feel happy with that. So, eventually,

you used to have the problem of not having a full scene [one of the 220 Amazon scenes], because it was not recorded, because the picture failed, because the antenna had a problem. Then, as you did not have a full Landsat scene, they were classified as 'not observed data.'" INPE coordinator. Personal communication.

As the technical failures have no relation to the topographic coverage of an area, the independence assumption holds for missing images. The relevance assumption also holds as Natura managers evaluated a municipality for entry on three criteria: the production potential of the location for a specific Amazon product, the organization level of the local supplier, and the information from INPE satellite data in the year before the entry. In the words of a Natura Ecorelations manager:

"We do not evaluate extensively whether the location had images for each specific year. But of course, we usually get the most recent available data [from the year before] and take a look to make the decision. For example, suppose it is a region that is very 'problematic' [a high number of missing images]. In that case, we might even decide to proceed with the relation in the new community, but aware of this problem." Natura Ecorelations coordinator. Personal communication.

The fact that Natura's Ecorelations team relies on prior-year satellite data also supports the use of the instrument through the relevance criterion. Therefore, the instrument also satisfies the relevance assumption, that is, the missing satellite images correlated with Natura's decisions about entry into municipalities.

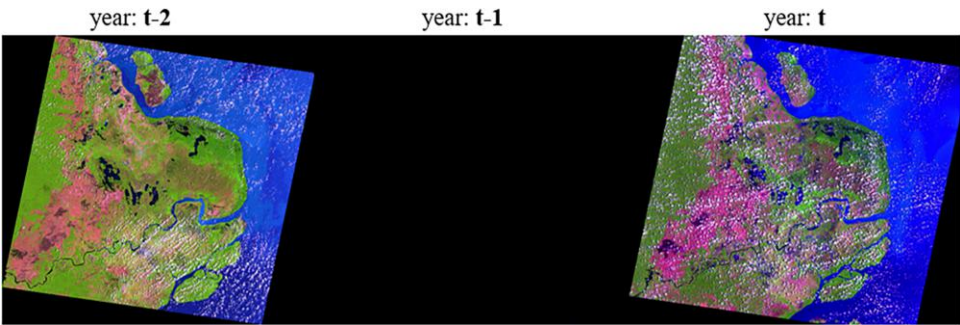
Although we cannot claim categorically that Natura was the only agent using the satellite data, it is unlikely that other agents, such as loggers and large farmers, used this information in their decisions to deforest an area. This is because technical failures seldom occurred in the same place in successive years. As a result, we claim that the instrument also satisfies the exclusion restriction assumption.

Considering that the current instrument satisfies all required assumptions, the paper presents the estimation of Natura's entry into municipalities on forest preservation and fire incidents through a just identified two-stage least squares (2SLS) DiD-IV as follows:

$$\begin{aligned} Natura_{m,t} &= \alpha' + \gamma'_m + \lambda'_t + \phi [\text{Log}(\text{Missing Image} + 1)_{m,t-1} \\ &\quad \times (1 - Natura_{m,t-1})] + X'_{m,t}\rho + \epsilon'_{m,t} \\ Y_{m,t} &= \alpha + \gamma_m + \lambda_t + \beta \widehat{Natura}_{m,t} + X'_{m,t}\rho + \epsilon_{m,t}. \end{aligned}$$

As the data set has a panel data structure, the instrument varies per municipality m and year t (Hudson et al. 2017). Therefore, it is essential to account for Natura's entry year into a municipality in the use of the instrument. For example, if Natura entered a municipality in year t , we would expect that managers used the information about missing images from year $t - 1$ in the decision making. However, in the years $t + 1$, $t + 2$, $t + 3$, and

Figure 4. (Color online) Illustration of “Missing Images” from Satellite Data



Source. Data from INPE PRODES. The illustration is provided for explanation purposes.
Notes. In the satellite images, the green colors represent forested areas, the red colors deforested or agricultural areas, the blue colors lakes, rivers, or sea, and the white colors clouds. In the year $t - 1$, this particular scene was not observed due to technical failures with the satellite.

onward, as Natura has already entered the municipality, its managers will not rely on satellite data in the decision to continue to interact with suppliers in the municipality. Thus, the models consider the term $(1 - Natura_{m,t-1})$ to account for the fact that the satellite information on missing images is used by Natura up to entry in year t but not subsequently.¹⁹

Results

The first stage results reported in Table 3, columns (2) and (4), show that Natura’s entry into a municipality was negatively correlated with the number of missing image areas ($p < 0.01$). The random incidence of a missing satellite in year $t - 1$ significantly diminished Natura’s propensity for entry into the municipality for which the missing image was associated in year t . The models reject the possibility of a weak instrument (weak identification) as the F statistic equals 12.46. These results reinforce the strong correlation between the instrument and the endogenous regressor, as well as the instrument’s adequacy (Mariano 1977, Angrist and Pischke 2008, Andrews et al. 2019).

In the second stage reported in Table 3, columns (1) and (3), the results correct for the endogeneity of Natura’s

entry decisions in assessing the significance of its impact. According to the models, conditional on the occurrences of missing satellite images in the first stage, Natura’s activity is significantly and positively associated with forest preservation ($p < 0.01$) in the second stage. However, Natura’s activity has no significance in analyzing fire occurrences. The magnitude of the coefficient for forest preservation equals 19.36%, which is considerably smaller than the estimate in the unadjusted TWFE DiD models.²⁰ Nevertheless, the economic significance of the result is still substantial. Considering this result, we estimate that Natura’s entry was associated with the preservation of 730,000 hectares of forest through the two decades, an area equivalent to 676,000 football (a.k.a. soccer) fields. In total, the firm avoided the emission of 58 million tons of carbon into the atmosphere.²¹ In other words, based on the DiD-IV estimation, the total preserved area associated with Natura’s entry into the Amazon is larger than the area of the state of Delaware during a period when the total loss of rainforest due to clear-cutting and burning is larger than the area of the state of California.

Unlike the unadjusted (TWFE DiD) models, the adjusted (DiD IV) models confirm that Natura’s activity is not

Table 3. Instrumental Variable Differences-in-Differences (DiD-IV) Analysis Using Satellite Missing Images

	(1) Second stage: Log (Forested Area + 1)	(2) First stage: Natura	(3) Second stage: Log (Fire Incidents + 1)	(4) First stage: Natura
Natura	19.36** (5.374)		0.362 (0.572)	
$\text{Log}(\text{Missing Image} + 1)_{t-1} \times (1 - Natura_{m,t-1})$		−0.0127** (0.00361)		−0.0127** (0.00361)
Year fixed effects	Yes	Yes	Yes	Yes
Municipality fixed effects	Yes	Yes	Yes	Yes
Satellite controls	Yes	Yes	Yes	Yes
R ²	0.552	0.704	0.963	0.704
Observations	13,676		13,676	
Number of municipalities	760		760	
Weak Identification test F-stat	12.46		12.46	

Note. Robust standard errors clustered at the municipality level in parentheses.
** $p < 0.01$; * $p < 0.05$; $^{\dagger}p < 0.1$.

associated with an increase in fire incidents in the locations in which the firm chooses to operate. The comparison of the two results suggests that Natura's entry selection may have been motivated by increasing fire pretrends, although more research is required to establish definitively whether the motivations were an intention for fire abatement.

Mechanisms Analyses

Three possible mechanisms are relevant to the results.

Mechanism 1: Tradeoff Between Conventional and Amazonian Agriculture

The main finding from the adjusted DiD-IV model is Natura's effect on forest preservation. How exactly did this preservation arise? Field evidence points to perceptions in Amazonian communities of variable benefits and costs to the communities of conventional versus Amazonian agriculture. As stated by a cocoa supplier:

"Usually, there is cocoa and pasture (...), but 70% of the people who are involved in the organic production program, the main income comes from cocoa. (...) if it wasn't for cocoa, you had to move cattle (...) it is as follows, to keep up with cattle, you need to have at least 100 hectares of pasture, right?! And with cocoa, a family with 10 hectares of cocoa, they already support themselves, got it? (...) these 10 hectares generate about R\$3,000 per month [US\$750]." Amazonian cocoa supplier. Personal communication.

The supplier describes that, if it were not for the cocoa program developed in partnership with Natura, the local rural producers would be compelled to switch from cocoa to cattle raising. To enable cattle production, these producers would have had to clear-cut the forest and transform the area into pasture land, a phenomenon documented by the literature (Alston et al. 2000, Pokorny et al. 2013, Rajão et al. 2020). In line with prior findings, this suggests that a possible mechanism by which Natura managed to promote forest preservation was by internalizing positive environmental externalities in providing a viable economic option to small rural producers to keep the forest standing while earning income from renewable products of the forest. This alternative increased forest preservation by aligning interests with preservation rather than deforestation. Therefore, fewer areas were converted to conventional agriculture such as cattle raising and soybean production.

The study uses data from the IBGE present on the Municipal Agriculture Production census database to test this argument. The IBGE database provides information on the yearly agricultural production of all Brazilian municipalities for various products. We selected six types of agricultural products on which to test the mechanism, including three conventional products (total soybean production area in hectares, total corn production area in hectares, and total cattle herd in number of animals) and

three Amazonian agricultural products (total açaí production area in hectares, total cocoa production area in hectares, and total passionfruit production area in hectares). "Conventional agriculture" refers to cattle farming and high-volume crops sold on competitive commodities markets, which can only occur on cleared land. "Amazonian agriculture" refers to renewable fruits, nuts, seeds, and berries originating in the Amazon biome and produced in a way that sustains the forest. The selection of the six products was based on two criteria. First, soybean, corn, and cattle are the main products of conventional agriculture in the Amazon area. Second, Natura has a long tradition of using açaí, cocoa, and passionfruit in its products.²²

We expect that Natura's entry into a municipality had a negative impact on sales of conventional agricultural products and a positive impact on sales of Amazonian agricultural products relative to the control group. To test for type II error, the models consider two "placebo" agricultural products for falsification tests. The first, a conventional product, is the total buffalo herd in number of animals. Buffalo raising is not a prominent alternative for deforested Amazon areas. The second, an Amazonian product, is total urucum production in hectares. Urucum is an Amazonian seed that Natura does not procure and for which it has no cosmetic line. The analysis examines whether Natura's activity is associated with higher levels of Amazonian products used in its cosmetics but not in urucum. Similarly, it examines whether Natura's activity is associated with lower levels of conventional agriculture prevalent in the area, but not buffalo.

Table 4 shows that Natura's activity is negatively and significantly associated with conventional agricultural products that are relevant in the area: soybean production (54.0% reduction at $p < 0.05$), cattle production (16.5% reduction at $p < 0.05$), and corn production (51.6% reduction at $p < 0.01$). Natura's activity had a null effect on buffalo production, which is the placebo for conventional agriculture. This validates that Natura's activity was associated with lower levels of conventional agriculture in the treated municipalities compared with the controls.

Table 5 shows that Natura's activity was positively and significantly associated with Amazonian agricultural products that are used as ingredients in its products: açaí production (166.0% increase at $p < 0.01$) and cocoa production (20.2% increase at $p < 0.05$). Surprisingly, passionfruit production is lower in areas of Natura's activity (36.1% reduction at $p < 0.01$). Natura's employees explained the reasons for this result. First, a potential tradeoff arose between the cultivation of açaí and passionfruit because both grow under similar conditions. Açaí's higher demand by Natura (almost three times higher than passionfruit) might have driven this result. Second, according to Natura's employees,

Table 4. Mechanism 1: Tradeoff Between Agricultural Products

	(1) <i>Log(Soybean Area + 1)</i>	(2) <i>Log (Cattle + 1)</i>	(3) <i>Log(Corn Area + 1)</i>	(4) <i>Log (Buffalo + 1)</i>
<i>Natura</i>	−0.540* (0.257)	−0.165* (0.0742)	−0.516** (0.177)	−0.204 (0.156)
Year fixed effects	Yes	Yes	Yes	Yes
Municipality fixed effects	Yes	Yes	Yes	Yes
Constant	1.229** (0.0857)	9.877** (0.0547)	6.151** (0.0497)	2.949** (0.0577)
Observations	14,440	14,440	14,440	14,440
<i>R</i> ²	0.841	0.918	0.784	0.802
Number of municipalities	760	760	760	760

Note. Robust standard errors clustered at the municipality level in parentheses.
***p* < 0.01; **p* < 0.05; +*p* < 0.1.

passionfruit can be cultivated in regions other than Amazon; therefore, the result might reflect a tendency to allocate resources toward açaí, which has higher “locational asset specificity” (Williamson 1996) to the Amazon region. Natura’s activity had a null effect on the placebo for Amazonian agriculture, that is, urucum production. Figure 5 represents these results visually.

In sum, the analysis of mechanism one provides evidence supporting the suggestion that the forest preservation promoted by Natura was associated with less conventional agriculture and more Amazonian agriculture in municipalities in which Natura was active as compared with those in which it was not. This analysis supports the assertion that, by using Amazon fruits, nuts, seeds, and berries as ingredients in cosmetics, shampoos, and soaps, Natura internalized the benefits of forest preservation in Coasean arrangements implemented in partnership with Indigenous communities and supported by the price premiums paid by ecologically oriented consumers.

Mechanism 2: Heterogeneity in Amazonian Products

The result on forest preservation shows the average effect that Natura had on communities. However, it is essential to consider the potential heterogeneous impacts of products in both forest preservation and fire incidents.

Some Amazonian agricultural products have a higher effect on forest preservation than others, particularly if, for instance, they depend on big Amazonian trees that are scattered around a large area. Additionally, a heterogeneous effect on fire incidents may also occur insofar as growing some of the products might trigger a higher number of small-scale fire occurrences, which happen at the forest edge and are not as harmful to the old forest (Silva et al. 2020).

Both possibilities are aligned with reports from Natura’s employees when describing a new cosmetic based on a sourced product:

“We are working with a new species now, the tucumã, which is a palm tree that grows mainly in pre-deforested areas. Similarly to babaçu, it grows in pasture areas (...); these species grow naturally in these areas trying to dominate the environment. They also grow inside the forest, but they are often found in these more open areas.” Natura Ecorelations coordinator. Personal communication.

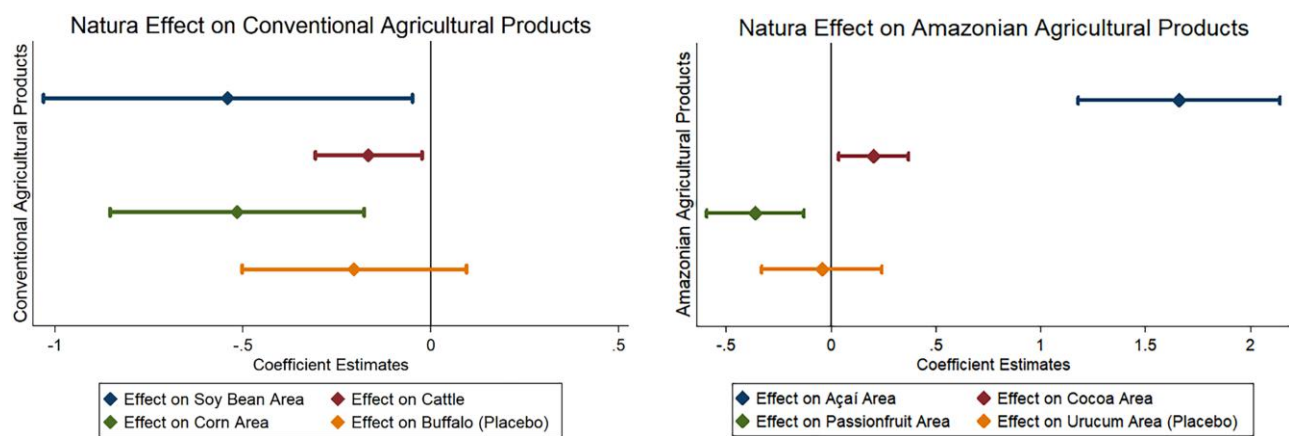
Natura’s effect on forest preservation might be the sum of its effect on both conservation and regeneration (Nunes et al. 2020).²³ To understand this possibility, the study presents a heterogeneity analysis of the impact that various Amazonian products had on forest preservation and the number of fire incidents. Table 6 presents these analyses.

Table 5. Mechanism 1: Tradeoff Between Agricultural Products

	(1) <i>Log(Açaí Area + 1)</i>	(2) <i>Log(Cocoa Area + 1)</i>	(3) <i>Log(Passionfruit Area + 1)</i>	(4) <i>Log(Urucum Area + 1)</i>
<i>Natura</i>	1.660** (0.251)	0.202* (0.0870)	−0.361** (0.120)	−0.0430 (0.150)
Year fixed effects	Yes	Yes	Yes	Yes
Municipality fixed effects	Yes	Yes	Yes	Yes
Constant	0.0281+ (0.0159)	0.813** (0.0284)	0.439** (0.0278)	0.249** (0.0237)
Observations	14,440	14,440	14,440	14,440
<i>R</i> ²	0.390	0.908	0.709	0.740
Number of municipalities	760	760	760	760

Note. Robust standard errors clustered at the municipality level in parentheses.
***p* < 0.01; **p* < 0.05; +*p* < 0.1.

Figure 5. (Color online) Mechanism 1: Tradeoff Between Agricultural Products



Notes. Data from the Brazilian Municipal Agricultural Production census. The graph’s x axes are on different scales.

Table 6. Mechanism 2: Amazon Products Heterogeneity Analysis

	(1) <i>Log(Forested Area + 1)</i>	(2) <i>Log(Fire Incidents + 1)</i>
Ucuúba (large-sized tree)	3.083** (0.294)	−0.198 (0.363)
Castanha (large-sized tree)	2.390** (0.486)	−0.0530 (0.398)
Andiroba (large-sized tree)	0.699** (0.260)	−1.178** (0.321)
Cacau (medium-sized tree)	0.982+ (0.509)	1.014** (0.390)
Pataqueira (small-sized herb)	1.035 (0.901)	3.232** (0.547)
Cumaru (large-sized tree)	0.0683 (0.462)	−1.540** (0.386)
Cupuaçu (medium-sized tree)	−0.00725 (0.468)	1.653** (0.470)
Açaí (large-sized palm)	−1.059** (0.224)	1.138** (0.249)
Patauá (medium-sized palm)	−1.073** (0.267)	0.347 (0.264)
Maracujá (small-sized bindweed)	−1.328* (0.591)	0.116 (0.633)
Tucumã (medium-sized palm)	−1.402** (0.289)	−0.888** (0.274)
Murumuru (medium-sized palm)	−1.558** (0.525)	−0.330 (0.451)
Priprioca (small-sized herb)	−2.223** (0.764)	−2.713** (0.552)
Babaçu (large-sized palm)	−5.984** (0.573)	−6.818** (0.389)
Year fixed effects	Yes	Yes
Constant	7.463** (0.459)	6.899** (0.366)
Observations	560	560
R^2	0.719	0.794
Joint significance test (F test)	94.95	294.61
F test (p value)	0.000	0.000

Notes. Robust standard errors in parentheses. All names are in Brazilian Portuguese. More details about each product can be found in Online Appendix 6.
** $p < 0.01$; * $p < 0.05$; + $p < 0.1$.

The results indicate considerable heterogeneity in forest preservation across the products. The joint significance of F tests equals 94.95 ($p < 0.000$). Estimates of individual coefficients indicate that three ingredients in the *Natura Ekos* line, namely ucuúba, Brazil nut (or castanha), and andiroba,²⁴ were strongly and positively associated with forest preservation, respectively, 308.3%, 239.0%, and 69.9% ($p < 0.01$). These fruits and nuts come from massive, old-growth Amazonian trees that are distributed throughout large areas of the forest and that are particularly associated with dense regions within the forest core. They grow slowly and are difficult to regenerate once destroyed. Trees in this area are often endangered and have high carbon mass, which makes them vulnerable to fire both from clear-cutting and from the spread of fire from nearby areas.

There is also heterogeneity in relationships between specific Amazonian agricultural products and fire incidents. The F test on differences in the coefficients equals 294.61 ($p < 0.000$) in this analysis. Products such as pataqueira, cupuaçu,²⁵ and açaí are positively and strongly correlated with fire incidents in their municipalities at, respectively, 323.2%, 165.3%, and 113.8% ($p < 0.01$). This occurs because these forest products (typically palm trees, other smaller trees, and herbs) may be traditionally cultivated using fire-based techniques on the understanding that reforestation is relatively quick given that they are mainly located at the forest edge. However, these traditional techniques make other forested areas, including old-growth forest, more vulnerable to the spread of fire. Figure 6 provides an illustration of the product heterogeneity analysis. Online Appendix 6 provides information on the biological differences between 14 products sourced by Natura. The results reinforce recent scientific arguments showing that “in Amazonia, fire typically occurs in forest edges by escaping from deforested areas, pastures, and agricultural fields and

leaking into surrounding forests” (Silva et al. 2020, p. 6). Cupuaçu and açaí grow on trees and palms that tend to be cultivated in former pasture lands, so they are associated with forest regeneration in the forest edge (Nunes et al. 2020, Silva et al. 2020). In contrast, pataqueira is a type of Amazonian grass (herb) that does not require much space and can be cultivated in irrigated beds. A blog post on Natura’s website reports that fire is traditionally used to produce this species.

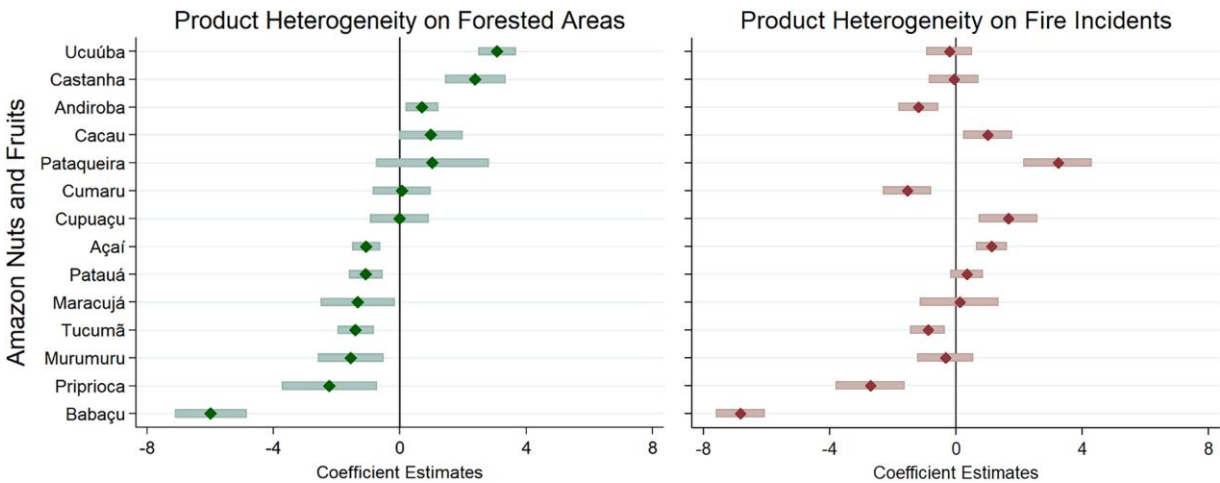
“According to local traditions, fire is used to plant pataqueira. The vegetation that margins the streams is burned and, after this, the plant is sown. We [Natura] had to find a new way of doing it without using fire, chemical inputs, and in a distant location from the water streams, which are protected areas. (...)

The combination of local knowledge and agronomy techniques showed results. Tests were made with irrigated beds, far away from the streams. We grew the plant in different ways simultaneously. Some techniques were proposed by us and others by the community in order to develop and define a combined growing system. This entire process was managed collectively, and through trial and error, new paths were discovered until we finally succeeded.” Available at the Natura Campus Blog (2016).²⁶

The quote illustrates why the pataqueira product has the strongest correlation with fire incidents among all products sourced by Natura, and yet the use of fire in sensitive areas carries dangers both for uncontrolled spread and environmental impact that had not been accounted for prior to Natura’s entry. The interviews suggest that, as a result, Natura aimed to work with communities to encourage the cultivation of pataqueira in ways that mitigated uncontrolled fire risk and river erosion.²⁷

The product heterogeneity analyses expose information on the relationship between Natura’s entry and forest preservation. The evidence suggests that Natura’s impact

Figure 6. (Color online) Mechanism 2: Amazon Products Heterogeneity Analysis



Notes. All names are in Brazilian Portuguese. More details about each product can be found in Online Appendix 6.

arose from both forest conservation in unharmed areas and forest regeneration at the forest edge. Regeneration occurred mainly in locations that had been previously deforested. Qualitative results aligned with the quantitative findings suggest that this may have occurred through Natura’s programs for rebuilding natural coverage by cultivating Amazonian products at the forest boundary. These results suggest that the seemingly paradoxical association between Natura’s entry and fire incidents may be related at least in part to Natura’s need for specific products that tend to grow at the forest edge, where small fire incidents occur in tandem with forest regeneration. Further research is required on the tension between Natura’s reforestation efforts and anthropic fire occurrences deliberately performed by large-scale ranchers who advocate for conventional agriculture.

Mechanism 3: Carbon Mass Effect

The third mechanism deals more deeply with the results of fire incidents from the unadjusted TWFE DiD model. The main analysis indicates that Natura’s entry is correlated with greater numbers of fire incidents by municipality. As suggested in the analysis of mechanism 2, this might arise from specific product choices such as pataqueira. Recall, however, that the pretreatment trends of fire incidents in control and treated municipalities differ. This difference suggests that Natura selects municipalities in which fire incidents are growing at unusually high rates. This suggestion is supported by the adjusted DiD-IV model and the placebo test (Online Appendix 7). The areas selected for entry by Natura have a higher carbon mass, and greater carbon mass is associated with larger numbers of fire incidents. The forested areas of the Amazon hold, on average, 188.5 tons of carbon per hectare (tC/ha). Deforested areas of the Amazon hold, on average, 107.9 tC/ha, so a forested Amazon hectare holds about 80 tC/ha more than a deforested one (Souza-Rodrigues 2019, p. 2730).

These findings and facts point to two factors as possible explanations for the higher number of fire incidents in municipalities chosen by Natura. First, the firm tended to enter municipalities with greater carbon density and thus greater fire risk. Second, after entry, Natura conserved and regenerated more forest than in the control group, which may have resulted in more fire incidents on the margin due to the enlarged carbon density of the local forest in the municipality. However, the qualitative and quantitative evidence suggests that this level of fire activity may have been relatively lower than would have occurred if Natura had not entered both because of Natura’s interventions to deter the use of fire as a cultivation technique at the forest edge and because of the incentives created by Natura’s relationships with Amazonian communities to cultivate fruit, nuts, and berries from both old-growth forest and reforested areas. A variable called forested proportion (in percent) was constructed to

examine evidence of this possibility. This variable measures the percentage of forested areas per total area in each municipality (e.g., more trees per square kilometer). We then evaluated the correlation between this variable with the number of fire incidents while considering whether Natura had entered. Table 7 presents the results.

Table 7, column (1), shows that the greater the forest proportion in a municipality, the greater the number of fire incidents ($p < 0.01$). This finding corroborates the previous interpretation of the carbon mass effect, that is, more trees per square kilometer are correlated with more fire incidents. Table 7, column (2), shows that, when controlling for Natura’s entry, the forest proportion coefficient remains significant ($p < 0.01$) and 3.6 p.p. greater in magnitude. This means that the number of fire incidents is even more strongly correlated with the proportion of forested areas in a municipality after Natura’s entry.

Overall, these results suggest that Natura enters municipalities with a greater forested area (i.e., relatively more trees per km² than other municipalities). As suggested, Natura’s entry may be correlated with more fire incidences because heavily forested municipalities are more likely to experience fire incidents. The results also point to the possibility that, upon entry, Natura contributed to increases in forest density by encouraging increases in the number of trees per square kilometer in the location even beyond the pre-entry level. This encouragement may have also led to increases in fire prevalence. Because Natura’s entry is associated with greater forest preservation, we conjecture that the higher level of fire occurrences after Natura’s entry may be of smaller scale, less devastating, and more carefully managed and contained than those in the control group (such as in the case of pataqueira). Better management of fire in the local communities in which Natura engages may result precisely from the Coasean

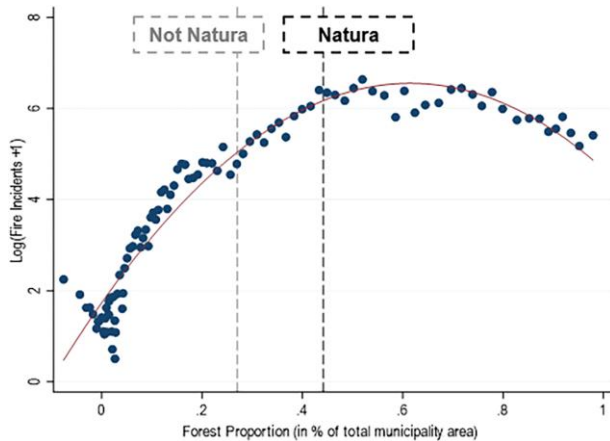
Table 7. Mechanism 3: Carbon Mass Effect

	Log(Fire Incidents + 1)	
	(1)	(2)
Natura × Forest Proportion		−2.129 (1.351)
Natura		1.991** (0.687)
Forest Proportion	4.699** (0.274)	4.735** (0.276)
Year fixed effects	Yes	Yes
Satellite controls	Yes	Yes
Constant	1.165** (0.123)	1.150** (0.123)
Observations	14,440	14,440
R ²	0.332	0.338

Notes. Robust standard errors clustered at the municipality level in parentheses. The forest proportion variable measures the percent of forested areas over the total area of a municipality.

** $p < 0.01$; * $p < 0.05$; + $p < 0.1$.

Figure 7. (Color online) Mechanism 3: Carbon Mass Analysis



Note. Dashed lines at the average forest proportion for “Not Natura” and “Natura” municipalities.

arrangements that align the interests of the firm and Indigenous communities to keep the forest standing because the forest is now an asset with direct economic value. If these associations hold up under further scrutiny, then they may explain at least in part the finding from the main TWFE DiD model that Natura’s entry is associated with more fire incidents. Figure 7 illustrates the analyses described in this section as mechanism three. Further research is needed to establish whether these relationships are robust.

Limitations and Future Research. In addition to the opportunities for further research listed previously, several other limitations in this phenomenological study raise opportunities for further research. The absence of micro-data on rules, norms, and governance structures within Amazonian rural associations and cooperatives hinders understanding the mechanisms behind the Coasean arrangements that we report on in this study. In particular, we do not have systematic evidence of the “polycentric governance” arrangements cited by prior researchers (Ostrom 1990, 2005; Gatignon and Capron 2023) as central to decision making in private orderings such as we observe between Natura and Indigenous communities in the Amazon. Data limitations also hampered the investigation into how Natura and its stakeholders developed their partnerships over time. Furthermore, the results point to the need for further theoretical developments to deepen conceptual links between (a) polycentric governance, (b) the new stakeholder management literature, and (c) measures of environmental impact. Specifically, this study suggests that decisions by suppliers to work with Natura require choices about mutual enfranchisement of stakeholders in polycentrism and that learning by Natura and stakeholders occurs over time. Together, these results point to greater

complexity in polycentric governance than has been previously theorized in the sense that suppliers may evolve in their capacities to negotiate, build coalitions, and engage in the graduated sanctioning that Ostrom (1990, 2005) and Gatignon and Capron (2023) identify as integral to polycentrism. Much more research is needed to understand these mechanisms and their implications.

Discussion and Conclusion

This phenomenological study describes the collaboration between Natura and its stakeholders to create an effective Coasean arrangement that aligns interests to internalize environmental externalities in the Amazon rainforest. Achieving remediation of climate change has been intractable partly because governmental interventions such as regulation, taxation, and coercive operations have not been fully effective. For corporate leaders, the challenges of aligning private-sector activity with ecological objectives have been similarly difficult to overcome. Internalizing environmental externalities and governing common-pool resources is particularly complex (Dowell et al. 2000, Henderson 2020). Specifically, in the Amazon region, deforestation through clear-cutting and burning for conventional agricultural production has had global consequences, demonstrating the intractability of addressing this commons-good problem for the government.

Yet despite the difficulties, this study provides evidence of success in achieving the kind of private ordering of stakeholder interests envisioned in Coase (1960). Natura, a well-known purpose-driven and stakeholder-oriented firm (Boehe et al. 2014, Lazzarini et al. 2020a, Marquis 2020, Serafeim 2022, Gatignon and Capron 2023), managed to cocreate a Coasean arrangement with environmentally conscientious consumers and rural Amazon communities. The alignment of interests among critical proximate stakeholders led to the achievement of both economic and environmental goals (Delmas and Toffel 2008; Eccles et al. 2014; Garcia-Castro and Aguilera 2015; Barney 2018; Klein et al. 2019; Bridoux and Stoelhorst 2020; McGahan 2021, 2023; Bacq and Aguilera 2022; Gatignon and Capron 2023) for the firm, its customers, and the supplying communities.

The analyses reported in this paper suggest that the environmental effects of Natura’s entry into the Amazon have global significance. Evidence shows that Natura contributed both to the deterrence of deforestation (i.e., to the remediation of negative externalities) and to the promotion of reforestation (i.e., to the achievement of positive externalities). Evidence in support of these claims is obtained through a series of econometric estimation models comparing municipalities into which Natura entered with municipalities that Natura did not enter. The findings suggest that Natura’s entry led to the preservation of an estimated 730,000 hectares, saving

approximately 58 million tons of Carbon (tC) between 2000 and 2018. Natura also contributed to shifts in crop production toward forest-sustaining alternatives and to increases in high-carbon forestation.

An analysis of mechanisms suggests that the increase in forestation occurred because Natura aligned the interests of Indigenous communities with that of forest preservation through the production of Amazonian agricultural products instead of conventional agricultural products. Heterogeneity in the products that Natura sourced led some communities to conserve and others to regenerate the forest. We also find evidence suggesting that Natura tended to enter areas vulnerable to fire incidence and then sought to foster reforestation and changes in local practices to reduce fire risk. Available evidence using missing images from satellites as an instrumental variable indicates that the positive association between Natura's entry and forest preservation was not accompanied by increased fire incidents.

This paper makes three main contributions to the management literature. First, the paper raises questions for theory by showing that stakeholder-oriented firms can contribute to preserving common goods such as the Amazonian rainforest through Coasean arrangements that align the interests with critical proximate stakeholders in complex, sustained ways that involve extensive institutional and relational investments. Creating and sustaining these arrangements requires much more than only focusing on transactions. Although much more research is needed to understand how complex Coasean arrangements arise and are sustained to address Grand Challenges such as climate change and rainforest preservation, the case of Natura provides a solid basis for optimism regarding the potential for impact of purpose-driven, stakeholder-oriented corporations.

Second, the paper advances empiricism through the analysis of a novel data set describing georeferenced satellite images to measure the impact of aligned stakeholder governance on environmental protection. Satellite data measures real environmental impact instead of relying on the firm's self-reported data or third-party rating scores. Objective assessments of the accomplishments of firms toward the achievement of critical social and environmental milestones of great public significance.

Third, the paper contributes to a practical understanding of how innovative managerial practices in the private sector foster the achievement of the UN's Sustainable Development Goals (SDGs), specifically SDG#13 (Climate Action) and SDG#15 (Life on Earth). By developing innovative approaches to stakeholder interest alignment, private-sector actors, and particularly leaders of multinational corporations, can take inspiration from Natura's achievements in preserving and reconstituting the Amazon rainforest.

There is no planet B.

Acknowledgments

Both authors contributed equally and are listed in alphabetical order of last name. The authors thank the special issue editors, co-editor George Serafeim, the associate editor, and the three reviewers for insightful comments and suggestions that contributed to the development of this study. The authors also thank Ruth Aguilera, Andy Back, Daphné Baldassari, Sandra Barbosa, Sandro Cabral, Laura Derksen, Aline Gatignon, Witold Henisz, Carlos Inoue, Sarah Kaplan, Nicola Lacetera, Sérgio Lazzarini, Bill McEvily, Mariana Oseguera Rodriguez, Patrick Rooney, Jieun Shin, Brian Silverman, Thomaz Teodorovicz, and participants in sessions at the Annual Social Entrepreneurship Conference, the Strategy and the Business Environment Conference, the Consortium on Competitiveness and Cooperation, the Alliance for Research on Corporate Sustainability Annual Conference, the Academy of Management Annual Meeting, the Strategic Management Society Annual Meeting, and Insper Metricis for valuable comments and suggestions. The authors thank Instituto Nacional de Pesquisas Espaciais), particularly Cláudio Almeida, and Natura Gerência de Relacionamento e Abastecimento da Sociobiodiversidade, especially Mauro Costa e Raoni Silva, for invaluable support. All remaining errors and omissions are our own responsibility. The University of Toronto Research Ethics Board's manager determined that this research is exempt from research ethics approval under TCPS2, Chapter 2, Article 2.1.

Endnotes

¹ PR Newswire. Available at <https://www.prnewswire.com/news-releases/natura-co-posts-strong-net-income-and-margin-improvement-in-q4-sales-grow-in-the-full-year-with-net-income-reaching-r1-billion-301499606.html>. Accessed October 5, 2022. Natura operates internationally. It acquired the Australian *Aesop* in 2014, the British *the Body Shop* in 2017, and the American *Avon* in 2019.

² Natura Ekos product line. <https://www.naturabrasil.fr/en-us/about-us/our-lines/ekos>. Accessed April 20, 2020.

³ UN Champion of the Earth 2015 video. Available at <https://www.youtube.com/watch?v=xdIctBWLIMandt=22s>. Accessed January 10, 2020.

⁴ Specifically, SDG 13.2 on integrating climate change measures into strategies and SDG 15.2 on promoting the implementation of sustainable management of forests.

⁵ A football (a.k.a. soccer) field's dimensions are 90 × 120 m.

⁶ To facilitate understanding of the economic rationale behind interest alignment, Online Appendix 1 presents a simple formal model based on Teodoridis (2018) that considers the choices of the Indigenous communities in interacting with Natura. Following the literature on value creation, appropriation, and distribution (Garcia-Castro and Aguilera 2015, Lieberman et al. 2017, Bacq and Aguilera 2022), the model suggests that, if the firm purchases specific forested products from farmers in Amazonian municipalities at prices above the next-best opportunities for these farmers, then these rural agents will seek to preserve the forest rather than to allow clear-cutting or burning. The challenge in implementing this Coasean arrangement for the firm is to assure that impediments to transactions are low enough to support collective action aligned with (i) Natura's ability to achieve a price premium on products generated from forest-preserving products, (ii) the creation of enough value through the sale of these products to support better economic returns for each enfranchised stakeholder than next-best alternatives, and (iii) the scope of authority of

stakeholders in each municipality to make decisions over how Amazonian land is productively deployed (i.e., over which crops are cultivated).

⁷ For anecdotal evidence of this phenomenon the documentary “Chico Mendes: Voice of the Amazon” (1989) and the movie “Iracema–Uma Transa Amazônica” (1975) provide an interesting picture of the social situation in the locality that is marked by deforestation, illegal trade of prime wood, slavery, prostitution, murder, and diverse forms of social marginalization and oppression. IMDB, available at <https://www.imdb.com/title/tt0293033/>. Accessed January 16, 2020.

⁸ Some municipalities might have one or more supplier cooperatives. Unfortunately, because of sample size limitations, we could not specify the effect of additional cooperatives in the same location, but we expect that the more partners Natura has in a given municipality, the higher should be their impact on forest conservation.

⁹ Expanding on Figure 1, Online Appendix 11 shows the composition of the total payments for two specific Amazonian products: açaí and Brazil nut. This online appendix was developed using aggregated data made available by Natura. The values for the two products are consistent with the averages described in Figure 1.

¹⁰ Online Appendix 12 presents the total investment Natura made in the Amazon rural communities from 2014 to 2019. Natura also contributed to the creation and development of the UEBT (The Union for Ethical BioTrade) certification, which respects the Nagoya Protocol on Access and Benefit-Sharing with Indigenous communities from using genetic resources and traditional pieces of knowledge of these groups. We believe this approach helps Natura build trustworthy relationships in their buyer-supplier relationships, reducing the transaction costs and allowing them to develop the Coasean mechanism connecting local Amazonian communities with environmentally concerned final consumers.

¹¹ UOL ECOA. Com investimento bilionário, Natura quer zerar desmatamento na Amazônia. Available at <https://www.uol.com.br/ecoa/ultimas-noticias/2020/07/13/com-investimento-bilionario-natura-quer-zerar-desmatamento-na-amazonia.htm?cmpid=copiaecola>. Accessed July 16, 2020.

¹² Economically, it was advantageous to the firm to access sustainable Amazonian products for the premium prices to final consumers. Socially, the legitimacy and effectiveness of the Amazonian operation depended on a close connection with local communities. Environmentally, the long-run sustainability of the firm’s business model depended on ecological conservation to the point that the top management team had part of their bonuses attached to the firm environmental performance.

¹³ The mean of forested areas is 3.995 km², the median equals 223 km², and the 75th percentile is 2.325 km². The number of fire incidents has a mean of 767 incidents, the median is 101 incidents, and the 75th percentile is 552 incidents. Removal of skew supports accuracy in the interpretation of the regression coefficients and consequently of the phenomena (please see distributions in Online Appendix 5). The dependent variables are expressed as $Y_{m,t} = \text{Log}(\text{Outcome}_{m,t} + 1)$ for each municipality m in each year t .

¹⁴ The study predictions were preregistered at the [AsPredicted.org](https://aspredicted.org) web page. The preregistration record can be accessed through the following link: <https://aspredicted.org/26wm7.pdf>.

¹⁵ Considering that the average forested hectare in the Amazon rainforest stores an additional 80 tons of carbon compared with a deforested hectare (Souza-Rodrigues 2019), Natura contributed to reducing about 145 million tons of carbon into the atmosphere. Online Appendix 13 presents estimates of the Natura’s impact on carbon tons under different scenarios considering the TWFE DiD and DiD IV estimations. It assesses the monetary value of the positive externality generated by Natura, and its suppliers if the carbon saved had been commercialized in the European carbon markets. In

the most conservative scenario this value would be of €728 million considering the TWFE DiD estimation (Figure 3(a)).

¹⁶ We performed several additional tests on the TWFE DiD models. Online Appendix 7 includes a placebo test simulating the results for the hypothetical entrance of Natura five years before the actual period of entry. As expected for a placebo test, the results are not statistically significant for forested areas but are for fire incidents. The coefficient of the fire incidents model column (4) is about 3.5 p.p. higher in the placebo test than in the main TWFE DiD model ($p < 0.01$). This finding confirms that the growing trend in fire incidents in treated municipalities preceded Natura’s entry. Moreover, it suggests that Natura selected to enter communities more vulnerable to fire. Online Appendix 8 shows that the results are robust to including both linear ($+t$) and quadratic ($+t^2$) time trends components. The magnitudes of the coefficients are similar to those in the TWFE DiD model. Online Appendix 9 reports whether pretreatment trends for treated and control municipalities are statistically similar. The results are robust for the forested area variable (no statistically significant difference in pretreatment trends) but not for the fire incidents variable ($p < 0.01$), reinforcing the evidence presented in Figure 3. Because Natura enters municipalities in different periods (staggered DiD), Online Appendix 10 describes a Bacon decomposition analysis of the results (Goodman-Bacon et al. 2022). The results show that 96% of the TWFE DiD estimator’s weight comes from the “treatedes versus never treated” group. This is plausible because just four municipalities are always treated over the study period, whereas 56 among 760 municipalities entered treatment gradually over the two decades under study. Therefore, these results confirm the adequacy of the TWFE DiD under this staggered situation. Finally, we also performed additional analysis using matching estimations both on demographics and distance to Natura industrial plant, with results that are robust with the main findings of the TWFE DiD. These last results have been omitted due to space limitations.

¹⁷ The TWFE-DiD models include municipality fixed effects that control for time-invariant unobservable characteristics at the municipality level. Nevertheless, this does not correct the selection Natura imposed when choosing the municipalities it entered.

¹⁸ Nagaraj (2022) used cloud cover as an instrument for firms’ investment decisions on the argument that cloudy images block visualization of terrains with potential for gold discovery and exploration. In this study, we cannot rely on cloud cover as an instrumental variable because a higher incidence of clouds is correlated with forest density and fire incidence, thus violating the independence criterion.

¹⁹ Online Appendix 14 presents an additional test of the instrumental variable estimation without the correction for Natura’s possible entrance in year $t - 1$. The results are statistically similar; however, in the latter case, we have a weakly identified model.

²⁰ As the 2SLS model has a level-log equation in the first stage and a log-level equation in the second stage, the interpretation of the β Natura in the second stage is already in percent.

²¹ Online Appendix 13 presents estimates of the Natura’s impact on carbon tons under different scenarios considering the TWFE DiD (Panel A) and DiD IV (Panel B) estimations. It assesses the monetary value of the positive externality generated by Natura, and its suppliers if the carbon saved had been commercialized in the European carbon markets. In the most conservative scenario this value would be of €294 millions considering the DiD IV estimation (Panel B).

²² Unfortunately, IBGE does not have information on other Amazonian products used by Natura.

²³ We define “conservation” as the maintenance of the native old forest. We define “regeneration” as the recovery of previously depleted and deforested areas. Thus, “preservation” is the aggregation of “conservation” and “regeneration.”

²⁴ See Natura Ekos: <https://www.naturabrasil.fr/en-us/about-us/our-lines/ekos/>. Accessed April 20, 2020.

²⁵ Natura Cupuaçu Video: <https://www.youtube.com/watch?v=I0ubNfwIEg>. Accessed April 20, 2020.

²⁶ Natura Campus Blog. Óleo essencial da pataqueira: união da técnica com as tradições amazônicas. Available at <http://www.naturacampus.com.br/cs/naturacampus/post/2016-04/oleo-essencial-pataqueira-tradicoes-amazonicas>. Accessed August 20, 2020. Translated by the author.

²⁷ An ideal analysis would have been to perform a triple-DiD comparing the (i) before versus after Natura, (ii) Natura municipalities versus non-Natura municipalities, and (iii) pataqueira municipalities versus non-pataqueira municipalities. Such an analysis would provide information about whether Natura's entry reduced the level of fire incidents in municipalities with pataqueira production before Natura's entry. Unfortunately, data are not available on pataqueira production in municipalities that did not enter, and thus we cannot proceed.

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**THERE IS NO PLANET B: ALIGNING STAKEHOLDER INTERESTS TO
PRESERVE THE AMAZON RAINFOREST,**

APPENDIX

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April 7, 2023

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APPENDIX 1 – FORMAL MODEL

Consider a set $V = \{V_{AMZ}, V_{CON}\}$ of opportunities to create value from agricultural products, where V_{AMZ} represents the local price of Amazonian agricultural production, such as *açaí*, Brazil nut, and cocoa. V_{CON} represents the global market price of conventional agricultural production, such as cattle, soybean, and corn. We define “Amazonian agriculture” as renewable fruits, nuts, and seeds originating in the Amazon biome and produced in a sustainable way. We define “conventional agriculture” as cattle farming and high-volume crops sold on competitive commodities markets, and which require clear-cutting of natural forests to occur. Consider a set $C = \{C_{AMZ}, C_{CON}\}$ of costs to produce different agricultural products, where C_{AMZ} is the cost incurred to produce or collect the Amazonian products, and C_{CON} is the cost incurred to deforest and produce conventional agricultural products. Finally, consider a set $B = \{B_1, \dots, B_n\}$ of value distributed to rural communities by the stakeholder-oriented firm. There are n types of possible ways by which value is distributed with the community. For instance, the stakeholder-oriented firm might distribute value through payments for traditional knowledge (aka. communitarian knowledge), investments in local infrastructure and institutions, provision of training and technical support, carbon credit purchases, and other techniques.

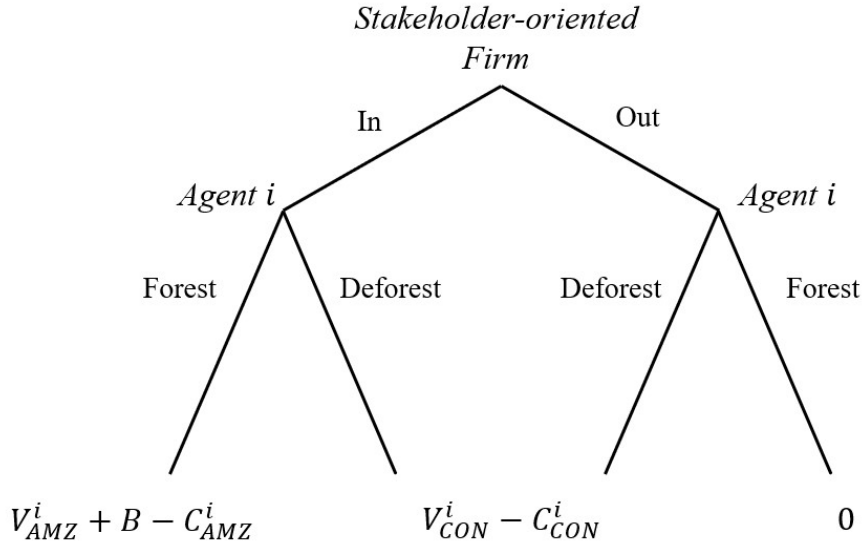
With these notations, consider a set of economic agents $i(V^i, C^i, B)$, each of which decides how to allocate effort in the production of alternative agricultural goods. V^i is the price of the production of one type of agriculture product by the agent i , C^i is the cost of the specific type of agricultural production chosen by i , and B is the total amount of value distributed to the community by the stakeholder-oriented firm. The model considers only two types of production: Amazonian agriculture (*AMZ*) and conventional agriculture (*CON*).

Assumptions. The model makes four main assumptions. First, it assumes, without loss of generality that $C_{AMZ}^i \equiv C_{CON}^i > 0, \forall i$. This means that the cost to produce an Amazonian agricultural product or a conventional agricultural product is equivalent and greater than zero for all agents. This assumption is motivated by the fact that to produce either type of product, agents expend commensurate time (given similar opportunity costs) and effort (given similar levels of technology available). Second, it assumes that $V_{CON}^i > V_{AMZ}^i \gg 0, \forall i$. This means that the global market price of conventional agriculture is always greater than the local price of Amazonian agriculture, and both prices are significantly greater than zero. Third, it assumes that the conventional agriculture market is perfectly competitive with perfect information. Note that assumptions two and three are motivated to represent the existence of global markets for conventional agriculture products.³ Amazonian agricultural products are rarely sold through open market exchange. Prior to the firm's entry, these products were traditionally harvested for community use and occasionally bartered between communities in closely negotiated and often non-monetary trades. These features reflect the reduced financial attractiveness of products based on Amazonian agriculture compared to those based on conventional agriculture prior to the firm's entry (Pokorny & Pacheco, 2014; Pokorny et al., 2013). Finally, the model assumes that the stakeholder-oriented firm maximizes profits when gaining access to Amazonian products. This arises because the firm commands a price premium when selling products incorporating Amazonian compounds to final ecologically conscientious consumers (Boehe et al., 2014).⁴

³ By "public markets for conventional products," we intend to describe markets with well-known operations and with financial trades that make information on prices and quantities more easily and quickly accessible to a broad range of participants and observers. In the particular case of the Amazon region, large trading companies – e.g., Archer-Daniels-Midland (ADM), Bunge, Cargill, and Louis Dreyfus – and meatpackers – e.g., JBS, Marfrig, and Minerva – are the main buyers in these markets.

⁴ As the firm always maximizes its payoff when accessing the Amazonian products, for simplicity, we are not including the firm's payoffs in here.

Agents' optimal choices. The model evaluates the optimal choices of each agent i as it seeks to maximize a payoff that depends on the presence, or not, of the stakeholder-oriented firm in the municipality where the agent's rural community is based. The decision tree below presents the agent i 's options and payoffs.



Decision Tree for Agent i in the Amazonian Rural Communities

Stakeholder-oriented Firm is Out. Consider the situation in which a stakeholder-oriented firm is **out** of the municipality where the agent's i rural community is located. The agent chooses to deforest or not the area in which the agent resides. If the agent opts to keep the **forest**, as there is no entity to which the Amazonian agriculture products may be sold, the potential value is not realized, but there is no cost incurred. In this case, the agent's payoff is zero. However, suppose the agent opts to **deforest** the area and convert it to conventional agriculture. In that case, the agent incurs a cost C_{CON}^i which equals the cost of deforestation plus production. At the same time, the agent receives the value V_{CON}^i when selling the conventional product in the market. Considering that the public market for conventional products is perfectly competitive, the agent's best response

is to *deforest* the area as long as the price of conventional products equals or exceeds the marginal cost of production.

Stakeholder-oriented Firm is In. Consider the situation in which the stakeholder-oriented firm is *in* the municipality where the agent's i rural community is located. Again, the agent chooses to deforest or not the area in which the agent resides. In this new scenario, if the agent opts to *deforest* the area and convert it to conventional agriculture, then the agent incurs a cost C_{CON}^i which equals the cost of deforestation plus production. At the same time, the agent receives the value V_{CON}^i when selling the conventional product in the market. However, suppose the agent opts to preserve the *forest*, as the stakeholder-oriented firm is present in the location. In this case, the agent may generate value by collecting and/or producing Amazonian products. Thus, the agent incurs a cost C_{AMZ}^i which equals the cost of collecting and/or producing these Amazonian products. At the same time, the agent received the price V_{AMZ}^i for the supply of these products to the firm. As the stakeholder-oriented firm distributes value to the municipalities, the agent's community receives an additional benefits payment B . Therefore, the agent's best response is to keep the *forest* standing as long as the additional benefits payments are greater than the difference between the prices from the conventional and Amazonian agriculture product types, i.e., $B > \Delta V = (V_{CON} - V_{AMZ}) \gg 0$.

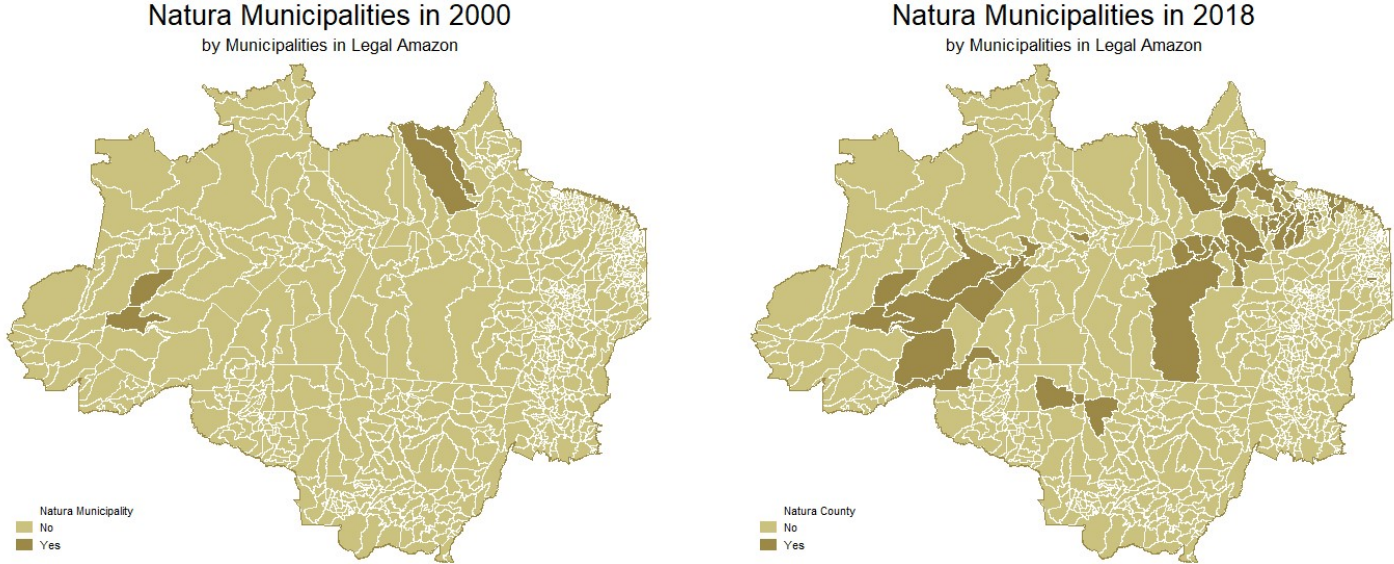
Proposition. *The equilibrium outcome depends on the additional benefits (B) paid by the firm once it enters a municipality, such that if the amount of additional benefits paid is sufficiently large, then, in equilibrium, agents will preserve the forest more than what might occur when the firm is not present.*

Proof. Consider the optimal choice of each agent i whose goal is to maximize the agent's payoff. It follows that when the value distributed by the firm is sufficiently large such that $B > \Delta V$, the sum of the local price of Amazonian agriculture (V_{AMZ}), and the additional benefits paid

by the firm (B) is larger than the global market price of conventional production (V_{CON}). Thus, if the firm enters a municipality, the agent has an economic incentive to preserve the forest that is greater than if the firm does not enter the municipality. Consequently, analyzing the subgame following history $h = In$, the subgame perfect equilibrium is $\Gamma(h) = (In, Forest)$. ■

Extension considering variations in the communal governance systems. Without loss of generality, we are assuming that the additional benefits payments (B) to each agent i within the same community are equivalent. We acknowledge that the communal governance of the rural association/cooperative in which agent i is a member can heterogeneously affect the payoffs of its members. To this end, we model that B would be subject to a parameter α , such that $\alpha \in [0,1]$. The parameter α measures the degree of equity in the communal governance/municipality. When $\alpha = 1$ we have a completely equitable communal governance in which B is shared evenly among all community members. This would be akin to Ostrom’s idea of a “polycentric governance” system (Ostrom, 1990, 2005). However, communal governance might alternatively involve lower levels of α , i.e. $\alpha < 1$, which would imply different levels of individual appropriation/capture over the total benefits paid (B) by the stakeholder-oriented firm with the community. We argue this is an opportunity for subsequent research, as described in the paper’s discussion section.

APPENDIX 2 – NATURA EVOLUTION IN BRAZILIAN AMAZON MUNICIPALITIES, FROM 2000 TO 2018



APPENDIX 3 – PICTURES FROM THE FIELD



Legend: Supplier's typical house (#1), *açaí* fruit shipment in a boat (#2), cocoa trees and fruits amid the forest (#3), and the Natura Industrial Plant at the city of Benevides (#4).

Note: All pictures were taken by the authors.

APPENDIX 4 – NATURA’S ADVERTISEMENT REINFORCING THE BENEFITS OF *MURUMURU* SOAP ON TECHNICAL, SOCIAL, AND ENVIRONMENTAL DIMENSIONS

Why

a *Mururu* Natura Ekos soap?



Because

it has a high concentration of *Mururu* butter, which has made it possible to create a flaky soap of abundant foam that protects your skin.

Because

Mururu butter is extracted by 157 families of producers from different communities in the Amazon whose extraction system conserves approximately 3,000 standing trees.

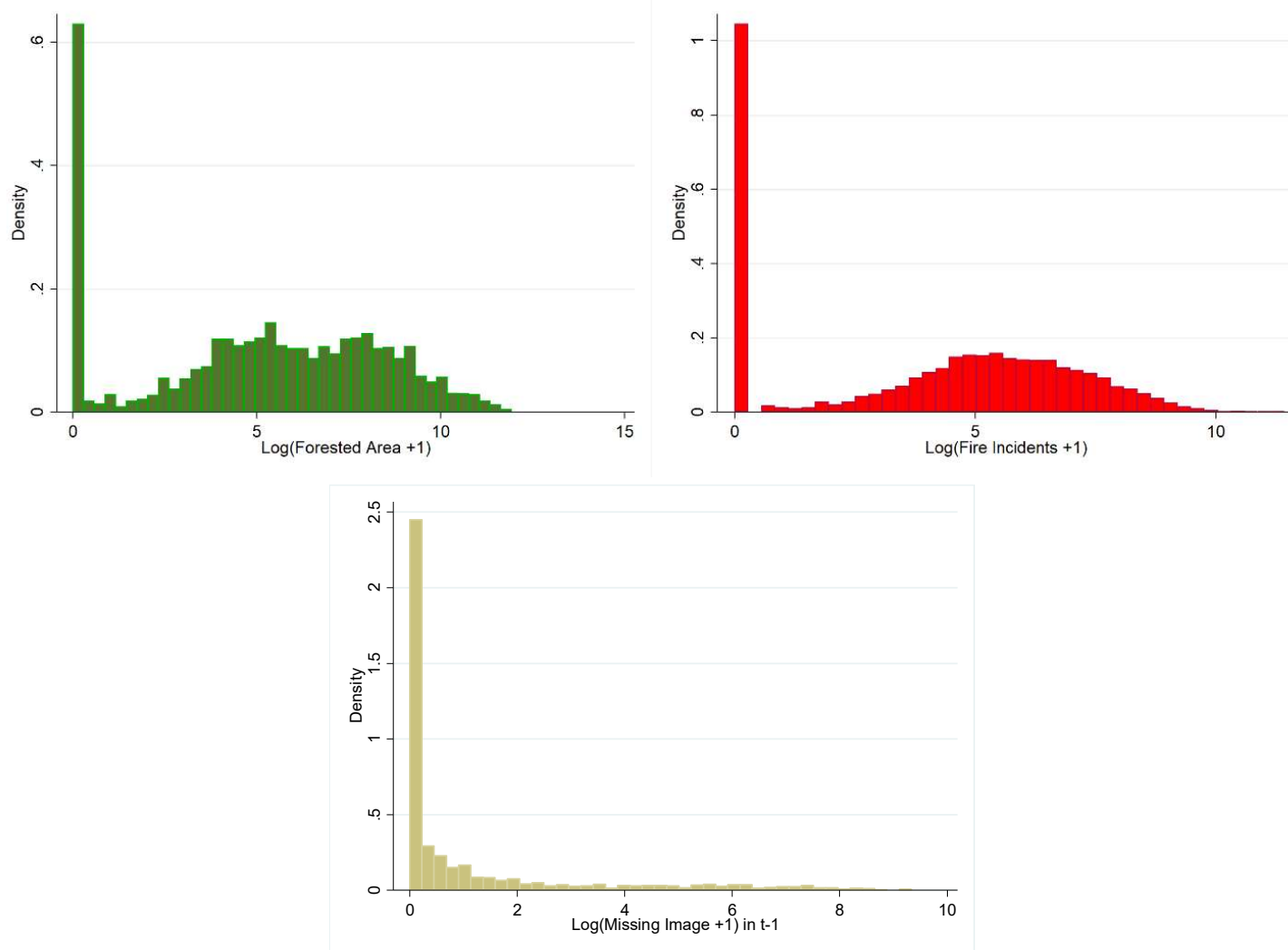


Because

resources return to communities valuing their work and keeping the forest alive.

Source: *Natura Ekos*. Translated by the authors.

APPENDIX 5 – HISTOGRAMS ON FOREST PRESERVATION INDICATORS AND INSTRUMENTAL VARIABLE



APPENDIX 6 – BIOLOGICAL DIFFERENCES AMONG NATURA’S SOURCED PRODUCTS

Popular Name	Scientific Name	Family	Ecological Succession Group	Type	Size	Density (number/hectare)	Production System	Conservation	Endangered
Ucuúba	<i>Virola surinamensis</i>	Myristicaceae	Mature	Tree	Large	10-17	Extractivism	Yes	Yes
Castanha (en. Brazil nut)	<i>Bertholletia excelsa</i>	Lecythidaceae	Mature	Tree	Large	15-20	Extractivism	Yes	Yes
Andiroba	<i>Carapa guianensis</i>	Meliaceae	Mature	Tree	Large	16	Extractivism	Yes	No
Cacau (en. Cocoa)	<i>Theobroma cacao L.</i>	Malvaceae	Mature	Tree	Medium	830	Cultivation	No	No
Pataqueira	<i>Conobea scoparioides</i>	Plantaginaceae	Herbfield	Herb	Small	N/A	Cultivation	No	No
Cumaru	<i>Dipteryx odorata</i>	Fabaceae	Climax	Tree	Large	5	Extractivism	Yes	No
Cupuaçu	<i>Theobroma grandiflorum</i>	Malvaceae	Mature	Tree	Medium	400	Cultivation	No	No
Açaí	<i>Euterpe oleracea</i>	Arecaceae	Mature	Palm	Large	200-400	Extractivism and Cultivation	Yes	No
Patauá	<i>Oenocarpus bataua</i>	Arecaceae	Climax	Palm	Medium	5	Extractivism	No	No
Maracujá (en. Passionfruit)	<i>Passiflora edulis</i>	Passifloraceae	Pioneer	Bindweed	Small	667	Cultivation	No	No
Tucumã	<i>Astrocaryum vulgare</i>	Arecaceae	Pioneer	Palm	Medium	50	Extractivism	No	No
Murumuru	<i>Astrocaryum murumuru</i>	Arecaceae	Climax	Palm	Medium	21-32	Extractivism	No	No
Pripioca	<i>Cyperus articulatus</i>	Cyperaceae	Herbfield	Herb	Small	N/A	Cultivation	No	No
Babaçu	<i>Attalea speciosa</i>	Arecaceae	Pioneer	Palm	Large	60-190	Extractivism and Cultivation	Yes	No

Note: the order of species follows Figure 6 and Table 6 in the main paper aiming to facilitate the reading. Data from multiple sources.

APPENDIX 7 – PLACEBO TEST ON NATURA’S ENTRANCE

DV =	(1) Log(Forested Area + 1)	(2) Log(Forested Area + 1)	(3) Log(Fire Incidents + 1)	(4) Log(Fire Incidents + 1)
Natura	0.261 (0.268)	0.314 (0.260)	0.565** (0.0998)	0.528** (0.0965)
Year FE	Y	Y	Y	Y
Municipality FE	Y	Y	Y	Y
Satellite Controls	N	Y	N	Y
Constant	5.231** (0.0378)	5.300** (0.0373)	2.559** (0.0444)	2.536** (0.0447)
Observations	14,440	14,440	14,440	14,440
R-squared	0.949	0.951	0.955	0.956
Number of municipalities	760	760	760	760

Note: Robust standard errors clustered at the municipality level in parentheses. As if Natura's entrance had happened five years earlier.

** p<0.01, * p<0.05, + p<0.1

APPENDIX 8 – NATURA EFFECT FOREST PRESERVATION INDICATORS CONTROLLING FOR TIME TRENDS

DV =	(1) Log(Forested Area + 1)	(2) Log(Forested Area + 1)	(3) Log(Fire Incidents + 1)	(4) Log(Fire Incidents + 1)
Natura	0.492* (0.222)	0.491* (0.223)	0.441** (0.0884)	0.442** (0.0888)
Year Linear Trend (+ <i>t</i>)	Y	Y	Y	Y
Year Quadratic Trend (+ <i>t</i> ²)	N	Y	N	Y
Municipality FE	Y	Y	Y	Y
Satellite Controls	Y	Y	Y	Y
Constant	-6.927 (7.058)	8,416** (974.0)	-127.8** (5.416)	-5,281** (880.7)
Observations	14,440	14,440	14,440	14,440
R-squared	0.950	0.950	0.939	0.939
Number of municipalities	760	760	760	760

Note: Robust standard errors clustered at the municipality level in parentheses.

** p<0.01, * p<0.05, + p<0.1

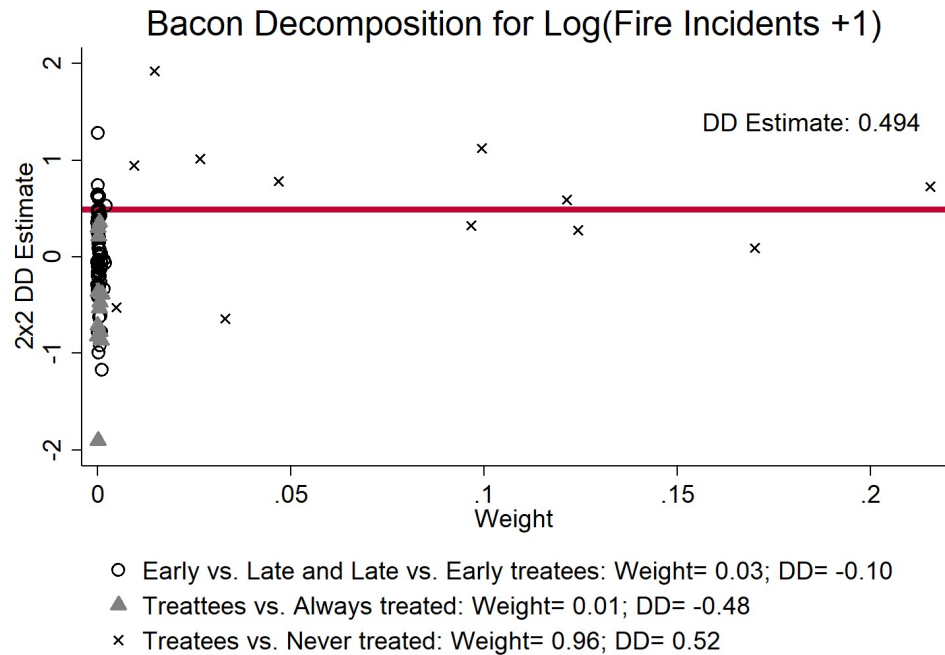
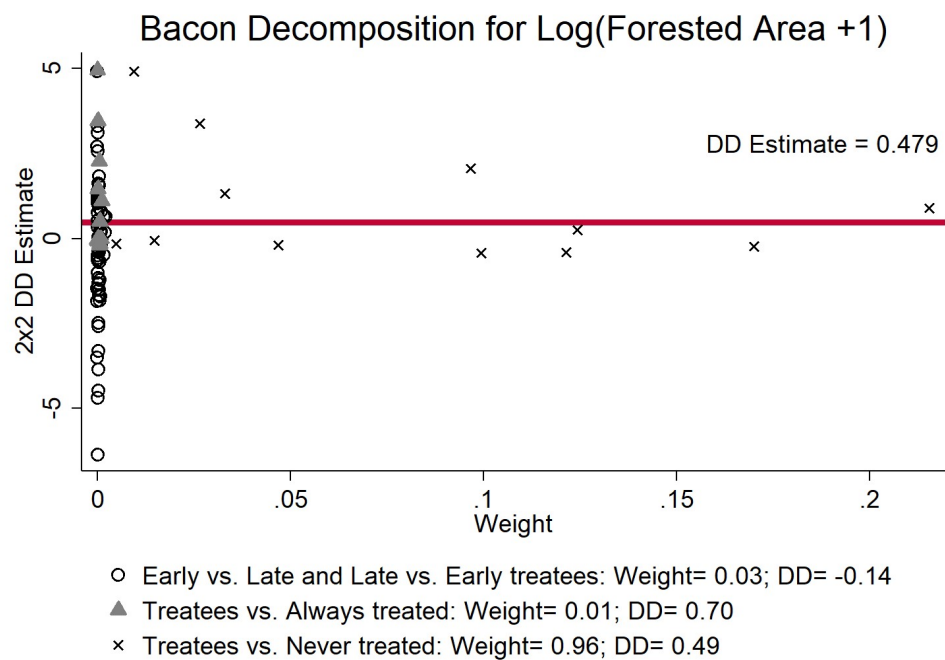
APPENDIX 9 – DIFFERENCES-IN-DIFFERENCES PRE-TREND TEST

DV =	(1) Log(Forested Area+1)	(2) Log(Forested Area+1)	(3) Log(Fire Incidents+1)	(4) Log(Fire Incidents+1)
Natura Treated Municipalities x Pretreatment Trend	0.0281 (0.0223)	0.0350 (0.0214)	0.0761** (0.0112)	0.0702** (0.0113)
Natura Treated Municipalities	-56.79 (44.91)	-70.67 (43.09)	-153.0** (22.46)	-141.2** (22.59)
Year Linear Trend (+ <i>t</i>)	0.00168 (0.00357)	-0.00137 (0.00346)	0.0933** (0.00352)	0.0935** (0.00351)
Year FE	Y	Y	Y	Y
Municipality FE	Y	Y	Y	Y
Satellite Controls	N	Y	N	Y
Constant	1.909 (7.176)	8.093 (6.963)	-183.9** (7.078)	-184.4** (7.055)
Observations	14,440	14,440	14,440	14,440
R-squared	0.949	0.951	0.956	0.956
Number of municipalities	760	760	760	760

Note: Robust standard errors clustered at the municipality level in parentheses.

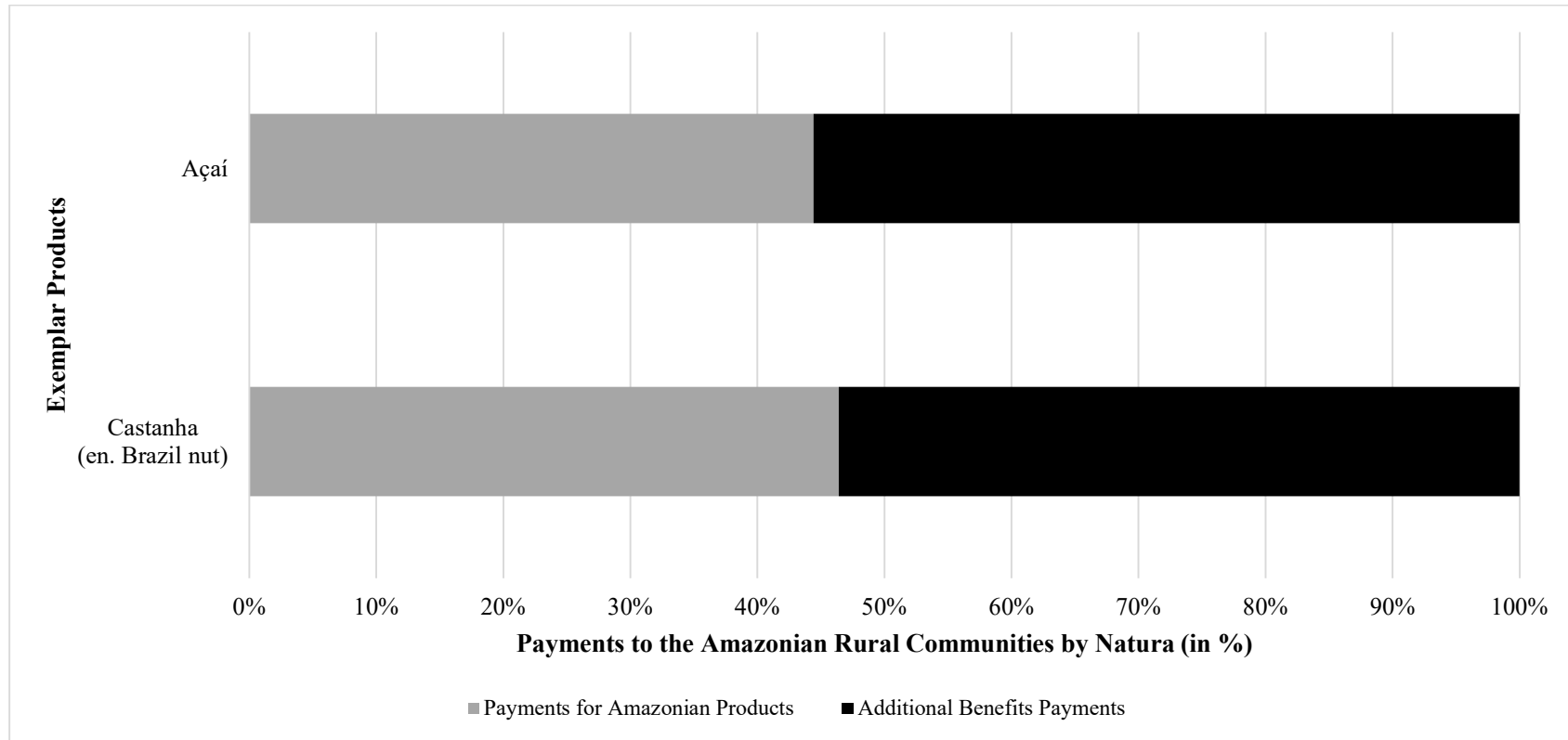
** p<0.01, * p<0.05, + p<0.1

APPENDIX 10 – BACON DECOMPOSITION OF TWO-WAY FIXED EFFECT DIFFERENCES-IN-DIFFERENCES ESTIMATOR



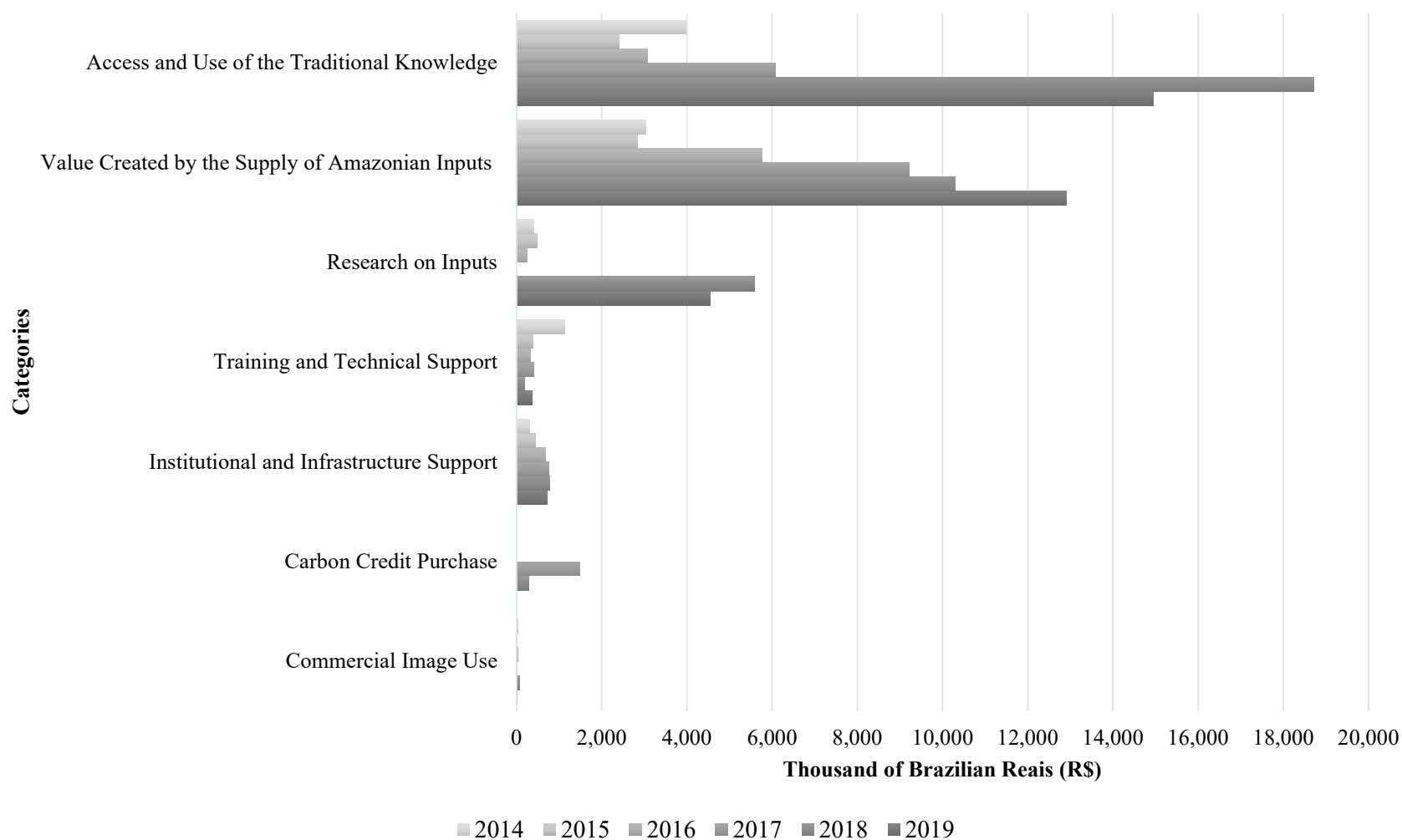
Note: based on Goodman-Bacon et al. (2019). Graphs y-axes are on different scales.

APPENDIX 11 – PAYMENT FOR AMAZONIAN PRODUCTS (V_{AMZ}) AND ADDITIONAL BENEFITS PAYMENTS (B) WITH THE AMAZONIAN COMMUNITIES BY NATURA, EXEMPLAR PRODUCTS



Note: Natura proprietary data. Natura managers shared microdata for both product payments from the year 2015 to 2020. We agreed with the firm managers to present the graphs on aggregate terms, given the data sensitivity for the firm business.

APPENDIX 12 – DETAILED VALUE DISTRIBUTED BY NATURA IN AMAZON COMMUNITIES FROM 2014 TO 2019



Note: Data from Natura's [Annual Report 2019](#) and [Annual Report 2016](#). GRI 203-1 indicator. Both were accessed on May 28th, 2020.

APPENDIX 13 – ESTIMATED EFFECT OF NATURA IN CARBON TONS SAVED AND POTENTIAL PRICE OF THE POSITIVE EXTERNALITY

Panel A: Considering TWFE DiD Estimation from Table 2, column (2)

Estimated Effect of Natura in Carbon Tons (tC) and Price (€)					
Carbon stock in Amazon Rainforest (tC/ha)	- 2sd	- 1sd	Mean	+ 1sd	+ 2sd
	26.4	53.5	80.6	107.7	134.8
Natura Estimated Preservation in Amazon Rainforest (in tC)	47,721,702	96,708,752	145,695,802	194,682,852	243,669,903
Natura Estimated Preservation in (in millions €)					
Price of Carbon under the European Union Emissions Trading System (€/tC)	€ 5	€ 239	€ 484	€ 728	€ 973
	€ 15	€ 716	€ 1,451	€ 2,185	€ 2,920
	€ 25	€ 1,193	€ 2,418	€ 3,642	€ 4,867
	€ 35	€ 1,670	€ 3,385	€ 5,099	€ 6,814
					€ 1,218
					€ 3,655
					€ 6,092
					€ 8,528

Note 1: Carbon stock in Amazon considers the difference between forested and deforested areas. Source (Souza-Rodrigues, 2018: 2730).

Note 2: Natura estimated preservation takes into account the total number of hectares “saved” by Natura (total of 1.807.640 hectares), which are based on the estimations presented in Table 2, column (2).

Note 3: Carbon prices set considering historical prices and the price on January 1st, 2020, which equals € 24.32/tC. Source: Bayer and Aklin (2020) and ICE EUA Futures. Available at: <<https://www.theice.com/products/197/EUA-Futures/data?marketId=5115271&span=3>> Accessed on April 16th, 2020.

Panel B: Considering DiD IV Estimation from Table 3, column (1)

Estimated Effect of Natura in Carbon Tons (tC) and Price (€)					
Carbon stock in Amazon Rainforest (tC/ha)	- 2sd	- 1sd	Mean	+ 1sd	+ 2sd
	26.4	53.5	80.6	107.7	134.8
Natura Estimated Preservation in Amazon Rainforest (in tC)	19,287,936	39,087,295	58,886,654	78,686,013	98,485,372
Natura Estimated Conservation in (millions €/tC)					
Price of Carbon under the European Union Emissions Trading System (€/tC)	€ 5	€ 96	€ 195	€ 294	€ 393
	€ 15	€ 289	€ 586	€ 883	€ 1,180
	€ 25	€ 482	€ 977	€ 1,472	€ 1,967
	€ 35	€ 675	€ 1,368	€ 2,061	€ 2,754
					€ 492
					€ 1,477
					€ 2,462
					€ 3,447

Note 1: Carbon stock in Amazon considers the difference between forested and deforested areas. Source (Souza-Rodrigues, 2018: 2730).

Note 2: Natura estimated preservation takes into account the total number of hectares “saved” by Natura (total of 730,604 hectares), which are based on the estimations presented in Table 3, column (1).

Note 3: Carbon prices set considering historical prices and the price on January 1st, 2020, which equals € 24.32/tC. Source: Bayer and Aklin (2020) and ICE EUA Futures. Available at: <<https://www.theice.com/products/197/EUA-Futures/data?marketId=5115271&span=3>> Accessed on April 16th, 2020.

APPENDIX 14 – INSTRUMENTAL VARIABLE DIFFERENCES-IN-DIFFERENCES (DiD-IV) ANALYSIS WITHOUT NATURA’S ENTRANCE CORRECTION

DV =	(1) Second Stage: Log (Forested Area+1)	(2) First Stage: Natura	(3) Second Stage: Log (Fire Incidents+1)	(4) First Stage: Natura
<i>Natura_t</i>	36.06* (14.38)		0.913 (1.023)	
<i>Log(Image not observed + 1)_{t-1}</i>		-0.00725* (0.00305)		-0.00725* (0.00305)
Year FE	Y	Y	Y	Y
Municipality FE	Y	Y	Y	Y
Satellite Controls	Y	Y	Y	Y
R-squared		0.700	0.963	0.700
Observations	13,676		13,676	
Number of municipalities	760		760	
Weak Identification test F-stat	5.66		5.66	

Note: Robust standard errors clustered at the municipality level in parentheses.

** p<0.01, * p<0.05, + p<0.1